

Math 137 Notes

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1 Lecture 1 (01/27/26)

We start by describing the origins of Algebraic Geometry. It all began in 1637 (so it's one year younger than Harvard). The motivation is that we try to relate the algebra of the polynomial to the geometry of the associated zero locus (points at which the polynomial is 0). Initially, people considered only the reals, and in affine space $(\mathbb{R}^n, \mathbb{C}^n)$. In the 19th century, Poncelet thought everyone was doing it wrong. He proposed that people should work over $\mathbb{C}$, and additionally should work over projective space (to be defined later). This simultaneously simplified everything, and made it far less intuitive.

Story time over: let's jump into definitions.

1.1 First Definitions

Definition 1.1. An *affine variety* is the zero locus in K^n of a collection of polynomials $f_\alpha \in K[x_1, \dots, x_n]$ (where K is a field).

In general, we will assume K algebraically closed, because it makes everything nicer. Recall:

$$\begin{aligned}\mathbb{P}_K^n &= \{\text{1-dimensional subspaces of } K^{n+1}\} \\ &= (K^{n+1} \setminus 0) / K^*\end{aligned}$$

Remark 1.2. It is important now to make a couple of notational remarks. We will often drop the reference to the field and just write $\mathbb{P}^n$ when the field is characteristic zero and algebraically closed. Feel free to pick the field as $\mathbb{C}$.

We will often write a point in $\mathbb{P}^n$ as $X = [X_0, \dots, X_n]$. Despite the notation, X_i is not a function of the point X . Due to our projectivization, $[X_0, \dots, X_n] = [\lambda X_0, \dots, \lambda X_n]$ for any nonzero λ . On the other hand, the ratios X_i/X_j are well-defined functions of the point X , as they are invariant under scaling of coordinates.

Observe that if $F(x_0, \dots, x_n)$ is a homogeneous polynomial of degree d , then

$$F(\lambda x_0, \dots, \lambda x_n) = \lambda^d F(x_0, \dots, x_n).$$

So, if we have a homogeneous polynomial on K^{n+1} , we can think of its zero locus as a subset of $\mathbb{P}^n$.

Definition 1.3. A *projective variety* (over K) is the zero locus of a collection of homogeneous polynomials $F_\alpha(Z_0, \dots, Z_n)$ of degree d_α .

We can find affine space living inside of projective space. Consider the following subset:

$$\mathbb{P}_K^n \supset U_0 := \{[Z_0, \dots, Z_n] \in \mathbb{P}_K^n : Z_0 \neq 0\} \cong K^n$$

with the isomorphism given by

$$[Z_0, \dots, Z_n] \leftrightarrow [Z_1/Z_0, \dots, Z_n/Z_0].$$

At this point, it is natural to wonder why it is useful to work in projective space. With this in mind, let's look at an example.

1.2 A Motivating Example

Example 1.4. Consider a *plane curve* $f \in \mathbb{R}[x, y]$ and its associated zero locus $C_f = \{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\}$.

For the sake of making our lives easier, assume that there does not exist a point (x, y) where f and both of its derivatives vanish. In other words, assume C_f is smooth.

Since C_f is a one dimensional closed manifold (due to the smoothness condition), we can ask how many connected components C_f has.

If f is degree 1, this is fairly simple as C_f is a single line.

If f is degree 2, there is already some complication. We can see C_f can be an ellipse or a parabola (with a single connected component) or a hyperbola (with two connected components).

If f is of higher degree, it becomes quite difficult to say how many connected components C_f will have.

Question 1.5. *In terms of $d = \deg(f)$, how many connected components can C_f have.*

Answer: the number of connected components of C_f is upper bounded by

$$\binom{d-1}{2} + 1 + \left\lfloor \frac{d}{2} \right\rfloor$$

and all numbers are achieved.

This is just the beginning! We can start to ask more complicated questions. How many of the connected components are lines or closed loops? Are they nested?

These questions are still very much open, despite the fact that these same questions were being asked hundreds of years ago. This should be somewhat scary and motivating for the introduction of projective space.

Let's look at the corresponding problem in complex projective space!

Example 1.6. Take $F(X, Y, Z)$ to be a homogeneous polynomial of degree d over $\mathbb{C}$ and define

$$C_F := \{[x, y, z] \in \mathbb{CP}^2 : F(X, Y, Z) = 0\}.$$

Again, let's make the assumption that C_F is smooth.

With these assumptions, C_F is a compact oriented 2-manifold. We can first ask for the number of connected components. We also would like to know the genus of each of the components. In contrast with the affine case, we can answer these questions definitively. We will prove (later in the course) that C_F is a connected manifold of genus $\binom{d-1}{2}$. Topologically, all of the zero loci of a degree d polynomial are the same! This is much better than the wild west of possibilities in the affine space.

1.3 Some First Varieties

Example 1.7. Take $\mathbb{P}_K^n$, as defined above. We can get $\mathbb{P}^{n-k} \subset \mathbb{P}^n$ as the zero locus of k independent homogeneous linear polynomials.

Definition 1.8. For any vector space V , we define the *projectivization*

$$\mathbb{P}V = \{1\text{-dimensional subspaces of } K^{n+1}\}$$

It will now be useful to have some facts. It is a good use of time to try to prove them in order to develop some familiarity with the definitions above.

Exercise 1.9. A single point is a projective variety.

Exercise 1.10. For any projective varieties X and Y , the union $X \cup Y$ is also a projective variety.

Combining the two previous exercises, we get that any finite set of points is in fact a projective variety.

Claim 1.11. Let $\Gamma = \{p_1, \dots, p_d\}$. Then Γ is the zero locus of a collection of homogeneous polynomials of degree $\leq d$.

In some ways, this result is sharp, as shown in the following exercise.

Exercise 1.12. Any homogeneous polynomial of degree strictly less than d that vanishes on d collinear points must vanish on the entire line containing those d points.

Is this the only example? If you want to go to the opposite extreme, you can ask: what if the points are in *linear general position*?

Definition 1.13. A set of points $\{p_1, \dots, p_d\} \subset \mathbb{CP}^n$ are in *linear general position* if any $n + 1$ of them span $\mathbb{C}^{n+1}$.

Question 1.14. What degree polynomials are needed to have a zero locus of d points in general position?

2 Lecture 2 (1/29/26)

Recall the definitions of affine and projective varieties from last lecture.

Definition 2.1. We define an ideal $\mathfrak{a}$ of a ring A to be a subset of A which is an additive subgroup and is such that $A\mathfrak{a} \subset \mathfrak{a}$. That is, $x \in \mathfrak{a}$ and $y \in \mathfrak{a}$ implies $xy \in \mathfrak{a}$ [Atiyah-Macdonald Commutative Algebra].

2.1 An algebraic aside

We have not, up to this point, specified whether the collection of polynomials defining a variety should be finite or infinite. In fact, it does not matter. This is due to the fact that polynomial rings in finitely many variables over a field are Noetherian. So, any ideal corresponding to a variety has finitely many generators.

2.2 Some basic examples of varieties

- Points/linear subspaces, both in the affine and projective settings. This can be seen by taking intersections of hyperplanes.
- Finite unions of linear subspaces, as shown by Theorem 1.10

We return to the question from last lecture: what degree polynomials do you need to define a collection of d points?

The naïve answer to this question: If Γ is a set of d points in $\mathbb{P}^n$, then Γ is the zero locus of polynomials of degree at most d .

Proof of Claim 1.11. Suppose you have $\{p_1, \dots, p_d\}$ and some $q \neq p_i$ for all i . For any i , we can find a hyperplane that contains p_i but not q (i.e. there exists a linear form L_i vanishing at p_i but not q). Let $F := \prod L_i$. $\square$

Sometimes the questions we ask are too easy. So, let's consider the same question of points in linear general position, as was posed at the end of last lecture. We can follow a similar protocol to the naïve algorithm above to show that Γ is the zero locus of polynomials of degree $\lceil d/n \rceil$. The details of this should be found on your problem set in the case $n = 2$.

Question 2.2. *Is this bound sharp?*

Question 2.3. *If we have 7 points in general position in $\mathbb{P}^3$ (i.e. no 4 points lie on a hyperplane), is Γ the locus of quadrics?*

This should give a taste of some of the problems we can ask with even fairly minimal definitions. In fact, questions about finite sets of points in $\mathbb{P}^n$ are the subject of current research.

2.3 Back to definitions, and some examples

Observe that the group PGL_{n+1} (invertible $(n+1) \times (n+1)$ matrices mod scalars) acts on $\mathbb{P}^n$.

Definition 2.4. We say varieties $X, Y \subset \mathbb{P}^n$ are *projectively equivalent* if there exists $A \in PGL_{n+1}$ such that $AX = Y$.

With this definition, we have the following fact (which is on the first problem set).

Exercise 2.5. Any two subsets of $\mathbb{P}^n$ of cardinality $n + 2$ in linear general position are projectively equivalent.

As an aside, for those familiar with some complex analysis, this implies the statement that any three points can be mapped to any other triple of points on the Riemann Sphere via a Möbius transformation.

Remark 2.6. One of the theorems we'll see later is the Lefschetz Principle. It says that any theorem in AG that can be proved for varieties over the complex numbers can be proven over any algebraically closed field of characteristic zero. This may seem somewhat powerful, but in fact we will be able to prove this fairly easily later in the course. So, if it brings comfort, feel free to assume that our fields are $\mathbb{C}$.

Definition 2.7. A variety $X \subset \mathbb{P}^n$ is a *hypersurface* if it is the zero locus of a single polynomial.

Now, let's introduce the first variety that is neither a hypersurface nor a linear subspace.

Definition 2.8. A *twisted cubic* curve $C \subset \mathbb{P}^3$ is the image of the map

$$\mathbb{P}^1 \rightarrow \mathbb{P}^3$$

given by

$$[X_0, X_1] \mapsto [X_0^3, X_0^2 X_1, X_0 X_1^2, X_1^3].$$

It is worth taking a moment to verify to oneself that this map is well defined. It is not immediately clear that the image of this map is in fact a variety in $\mathbb{P}^3$.

Claim 2.9. C is the zero locus of the three homogeneous quadratics $Z_0 Z_2 - Z_1^2$, $Z_0 Z_3 - Z_1 Z_2$, and $Z_1 Z_3 - Z_2^2$.

The proof of this claim is left to the problem set.

There are some questions which arise immediately. For example, do we need all three equations? This is, once again, is on the homework. To see why you need all three equations, consider

Claim 2.10. C is the locus in $\mathbb{P}^3$ of $[Z_0, \dots, Z_3]$ such that

$$rk \begin{pmatrix} Z_0 & Z_1 & Z_2 \\ Z_1 & Z_2 & Z_3 \end{pmatrix} \leq 1$$

This variety answers a previous question, by taking any seven points on the twisted cubic. Observe that any such points will be in linear general position, and that any quadratic polynomial vanishing at these points vanishes identically on the twisted cubic. In particular, this shows that the ideal of seven points cannot be generated by quadrics.

This argument can be generalized to higher dimensions

Definition 2.11. A rational normal curve $C \subset \mathbb{P}^d$ is the image of the map $\mathbb{P}^1 \rightarrow \mathbb{P}^d$ sending

$$[X_0, X_1] \mapsto [X_0^d, X_0^{d-1} X_1, \dots, X_1^d].$$

Again, we can verify that this is a well-defined map (of sets) and that C is a variety via a very similar argument to the one used for the twisted cubic.

We have one more generalization.

Definition 2.12 (Veronese map). We have a map $\mathbb{P}^n \rightarrow \mathbb{P}^N$ (with $N > n$) which sends

$$[Z_0, \dots, Z_n] \mapsto [\dots, Z^I, \dots]$$

where for a tuple $I = (i_0, \dots, i_n)$ with $\sum i_\alpha = d$, $Z^I = \prod Z_\alpha^{i_\alpha}$. The image is called the *Veronese variety*.

The fact that this is in fact a variety is somewhat messy, but is a special case of results from next week about morphisms between varieties.

One last example before we move on.

Definition 2.13. A *plane conic curve*, $V(Q)$ is the zero locus of a homogeneous of a quadratic polynomial $aZ_0^2 + bZ_1^2 + cZ_2^2 + dZ_0Z_1 + eZ_0Z_2 + fZ_1Z_2$ in $\mathbb{P}^2$.

Remark 2.14. There are some degenerate cases when this quadratic factors, possibly even as the square of a linear polynomial.

Now, we note that the space of plane conic curves is exactly parameterized by $\mathbb{P}^5$. This is a common phenomenon in AG, where families of our objects are parameterized by points in a different variety.

We will get started next time with morphisms between varieties and the Zariski Topology.

Definition 2.15 (Zariski topology). Say $X \subset \mathbb{P}^n$ is a projective variety. The *Zariski topology* on X is defined by saying the closed sets $Z \subset X$ are exactly the subvarieties of X .

Exercise 2.16. It is good to verify that the Zariski topology defined above is in fact a valid topology.

As a warning, it is not a good idea to think about the Zariski topology as a topology in the conventional sets.

Take, for example $\mathbb{P}^1$. The closed subsets are then $\mathbb{P}^1$ and finite subsets. Then, the open subsets are exactly the empty set and the cofinite sets. This means that every bijection is a continuous map. This is somewhat unreasonable, but it is in fact what we will use. So, be careful going forward when we talk about open sets.

3 Lecture 3 (2/3/26)

Now that we have defined varieties, the natural next step is to define morphisms between two varieties. As we discussed at the end of lecture last time, in order to do this, we need to introduce the Zariski topology. Recall the definition of the Zariski topology from above.

Remark 3.1. This “topology” provides incredibly helpful language when we are talking about varieties. Going forward, when we say open or closed set, we mean in the Zariski topology. It is very useful for a number of reasons. First, this topology is independent of a choice of field, so we do not need to rely on inheriting the topology from the field. When we work over $\mathbb{C}$, we inherit a topology for $\mathbb{C}^k$ or $\mathbb{P}\mathbb{C}^k$. We will call the inherited topology the *classical topology*.

Remark 3.2. Some Zariski topology facts:

- The Zariski topology is **Noetherian**. Meaning, if we have

$$X \supset Y_1 \supset Y_2 \supset \dots$$

where the Y_i are closed, then $\exists m$ such that the descending chain

$$Y_m = Y_{m+1} = Y_{m+2} = \dots$$

stabilizes. This is equivalent to the fact that polynomial rings are Noetherian.

- If X is a top. space, we say $U \subset X$ is locally closed if $\forall p \in U, \exists$ a neighborhood $p \in V \subset X$ such that $U \cap V$ is closed in V . Equivalently, U is the intersection of a closed set and an open set, an open subset of a closed subset of X , or a closed subset of an open subset of X . (It is a good exercise to verify that all of these notions are equivalent).

Definition 3.3. A *quasi-projective variety* is a locally closed subset of $\mathbb{A}^n$ or $\mathbb{P}^n$.

Remark 3.4. The above definition includes both affine and projective varieties. Also note that we can drop the $\mathbb{A}^n$ from the definition, as all locally closed subsets of $\mathbb{A}^n$ are locally closed subsets of $\mathbb{P}^n$.

Example 3.5. Here is one useful example (and corresponding non-example):

- $\Gamma = \{(x, y) : y = 0, x \neq 0\}$ is quasi-projective.
- $\mathbb{A}^2 \setminus \Gamma = \Gamma^c = \{(x, y) : y \neq 0 \text{ or } x = y = 0\}$ is not a quasi-projective variety. This can be seen by looking locally near the origin.

3.1 Regular maps

We now want to define morphisms between varieties. In order to do this, we need regular functions. We will start with the affine case and then proceed to the more general case.

Let's start with $X \subset \mathbb{A}^n$ affine variety and the ideal of functions vanishing on X

$$I(X) = \{f \in K[z_1, \dots, z_n] \mid f|_X \equiv 0\}$$

Definition 3.6. The *coordinate ring* of X is defined as

$$A(X) = K[z_1, \dots, z_n]/I(X)$$

We would like regular functions to look like the restriction of a polynomial in $z_1, \dots, z_n$ to X , modulo those vanishing on X . Say $U \subset X$ open.

Definition 3.7 (Regular functions). A function f on U is *regular* if for all $p \in U$, there exists a neighborhood $V \subset U$ of p such that $f|_V = g/h$ for $g, h \in A(X)$ and $h(p) \neq 0$.

So, if $X \subset \mathbb{A}^n$ is closed (is an affine variety), then any element of $A(X)$ is a regular function on X . A priori, it is not clear that these are the only regular functions on all of X . But, this turns out to be the case.

Claim 3.8. If $X \subset \mathbb{A}^n$ is closed, then the ring of regular functions on X is $A(X)$.

We will defer the proof until we have the Nullstellensatz at our disposal (Chapter 5 of Harris's book).

Definition 3.9 (Distinguished opens). If $X \subset \mathbb{A}^n$ is closed, we say the open subsets $U_f = \{p \in X \mid f(p) \neq 0\}$ are *distinguished open sets*.

Claim 3.10. The regular functions on U_f are $A(X)[\frac{1}{f}]$

This claim is once again a consequence of the Nullstellensatz.

Now, let's take a closed subset $X \subset \mathbb{P}^n$ (so X is a projective variety). Now, X is covered by affine varieties, namely the sets $U_0, \dots, U_n$ where

$$U_i := \{Z_i \neq 0\} \cong \mathbb{A}^n$$

Then, $X \cap U_i$ is an affine variety.

Definition 3.11. Say a function f on $U \subset X$ is regular if $f|_{U \cap U_i}$ is regular for all i .

Remark 3.12. There is often a simpler way to express regular functions on $X \subset \mathbb{P}^n$. For example, we can take a ratio

$$\frac{F(Z_0, \dots, Z_n)}{G(Z_0, \dots, Z_n)}$$

where $\deg F = \deg G$ and G is nowhere 0 on X .

3.2 Morphisms

Again, we will start with the simpler affine case. Let X be a quasi-projective variety.

Definition 3.13 (Affine regular maps). A map $\phi : X \rightarrow \mathbb{A}^n$ is *regular* if each coordinate function is regular, i.e.

$$\phi(p) = (f_1(p), \dots, f_n(p))$$

for regular f_i .

If $Y \subset \mathbb{A}^n$ is a variety, a regular map $X \rightarrow Y$ is a regular map $f : X \rightarrow \mathbb{A}^n$ with $f(X) \subset Y$.

Definition 3.14 (Projective regular maps). Similarly, a map $\phi : X \rightarrow \mathbb{P}^n$ is *regular* if $\forall i = 0, \dots, n$ the restriction

$$\phi_i = \phi|_{\phi^{-1}(U_i)} : \phi^{-1}(U_i) \rightarrow \mathbb{A}^n$$

is regular.

Remark 3.15. If $X \subset \mathbb{P}^n$ is quasi-projective, then we can sometimes write

$$\phi : X \rightarrow \mathbb{P}^n$$

as

$$\phi[X_0, \dots, X_m] = [F_0(X), \dots, F_n(X)]$$

where the F_i are all of the same degree and not all 0 at any $p \in X$. Any map that can be represented in the form is regular. But, not all regular maps can be written in this way.

Let's do one example to finish out this lecture.

Example 3.16 (Veronese map). Let $X = \mathbb{P}^m$ with homogeneous coordinates $X_0, \dots, X_m$. Fix a degree d , and consider the map $\mathbb{P}^m \rightarrow \mathbb{P}^N$ given by $[X_0, \dots, X_m] \mapsto [\dots, X^I, \dots]$ where $X^I = X_0^{i_0} \dots X_m^{i_m}$ such that $\sum i_\alpha = d$. This is a more general form of the rational normal curve (in that case, $m = 1$).

4 Lecture 4 (2/5/26)

Recall from Lecture 2 (Definition 2.12) the definition of the Veronese map.

Example 4.1. The image of the Veronese map $V_{2,2} : \mathbb{P}^2 \rightarrow \mathbb{P}^5$ is called the Veronese surface.

Proposition 4.2. *The image of $V_{d,n}$ is a projective variety of $\mathbb{P}^N$.*

Proof Sketch. Let $I = (i_0, \dots, i_n)$ with $\sum i_j = d$. Define J, K, L symmetrically with $I + J = K + L$. Check that $V_{d,n}$ is the zero locus of the quadratic polynomials $W_I W_J - W_K W_L$. $\square$

Let's look at some consequences (insignificant according to Prof. Harris):

Take any projective variety $X \subset \mathbb{P}^n$. We can write $X = V(F_1, \dots, F_m)$ with F_i homogeneous of degree d_i . Choose $d > \max(d_i)$. We can then write $X = V(G_1, \dots, G_k)$ where the degree of G_i equals d by taking F_i times all of the monomials of degree $d - d_i$.

Conclusions:

1. Any variety $X \subset \mathbb{P}^n$ is isomorphic to the intersection of $V_{d,n}$ with a linear subspace of $\mathbb{P}^n$. This can be seen by writing X as the common zero locus of degree d polynomials, and then pulling back via the Veronese map.
2. Any variety is isomorphic to an intersection of quadrics.

The second point seems very cool, but is in fact completely useless (according to Professor Harris).

Now, looking forward, we will eventually prove the following key result.

Theorem 4.3. *The image of a regular map from a projective variety into a projective space is a variety.*

It is good to verify that the equivalent statement is not true in the affine setting.

Exercise 4.4. There exists a regular map $f : X \rightarrow \mathbb{A}^n$ with X an affine variety such that $f(X)$ is not an affine variety.

It is not useful to have another useful construction.

Definition 4.5. The *segre map* is defined as follows:

$$\sigma_{m,n} : \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^{mn+m+n}$$

$$([X_0, \dots, X_m], [Y_0, \dots, Y_n]) \mapsto [\dots, X_i Y_j, \dots].$$

Remark 4.6. As always, you should verify to yourself that the map is well defined. Also, we can verify that the image is a projective variety, called the Segre variety $\Sigma_{m,n}$ as it is the zero locus of polynomials $Z_{ij} Z_{kl} - Z_{il} Z_{kj}$.

Let's look at some specific examples of Segre maps.

Example 4.7. The simplest example $\sigma_{1,1} : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3$ maps

$$([X_0, X_1], [Y_0, Y_1]) \mapsto [X_0 Y_0, X_0 Y_1, X_1 Y_0, X_1 Y_1]$$

and we'll call these coordinates $Z_0, \dots, Z_3$ respectively. The image is $Z_0 Z_3 - Z_1 Z_2 = 0$.

We can write this variety in the form

$$\det \begin{bmatrix} Z_0 & Z_1 \\ Z_2 & Z_3 \end{bmatrix} = 0.$$

Remark 4.8. A priori, the segre embedding is not a regular map, as $\mathbb{P}^m \times \mathbb{P}^n$ is not a projective variety. So, the segre map is just a map of sets until we show that $\mathbb{P}^m \times \mathbb{P}^n$ is isomorphic to a projective variety.

In general, we can say that a projective subvariety of $\Sigma_{m,n} = \mathbb{P}^m \times \mathbb{P}^n$ is the zero locus of a collection of bihomogeneous polynomials of arbitrary bidegree.

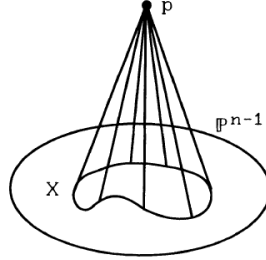
Definition 4.9 (Bihomogeneity). We say a polynomial in $k[X_1, \dots, X_m, Y_1, \dots, Y_n]$ is **bi-homogeneous** if it is homogeneous as a polynomial in $(k[X_1, \dots, X_m])[Y_1, \dots, Y_n]$ and $(k[Y_1, \dots, Y_n])[X_1, \dots, X_m]$.

We have one more construction (cones) before we get into the core problem of projections.

To construct cones, consider the following objects in $\mathbb{P}^n$. We have a **hyperplane** $H \cong \mathbb{P}^{n-1} \subset \mathbb{P}^n$, and $p \in \mathbb{P}^n \setminus H$. For example, in terms of coordinates $Z_0, \dots, Z_n \in \mathbb{P}^n$, we can choose $H = V(Z_n), p = [0, \dots, 0, 1]$. Suppose now we have a projective variety $X \subset H$.

Definition 4.10 (Cones). The **cone over X given $p, \overline{X, p}$** is defined to be

$$\bigcup_{q \in X} \overline{pq}$$



This is also a projective variety. If X is the zero locus of polynomials $\{F_\alpha\}$, where F_α are in n variables, $\overline{X, p}$ is the zero locus of these same polynomials viewed in $n+1$ variables. This is a specific example of a more general definition, called the **join** of two varieties, that we'll talk more about later.

Definition 4.11 (The join of two disjoint varieties). Let $X, Y \subset \mathbb{P}^n$ be projective varieties. We define the **join** of X and Y to be Zariski closure of the union of all lines in $\mathbb{P}^n$ which intersect X and Y .

We're now going to look at a similar construction: projection maps. Let $H \cong \mathbb{P}^{n-1} \subset \mathbb{P}^n$ and $p \in \mathbb{P}^n \setminus H$ be as before. Suppose now that $X \subset \mathbb{P}^n$ is a subvariety such that $p \notin X$, but

X not necessarily contained in H . We can define a map

$$\begin{aligned} X &\xrightarrow{\pi} H \\ q &\mapsto \overline{pq} \cap H \end{aligned}$$

i.e. we map each point $q \in X$ to the intersection of the line $\overline{pq}$ and H .

Claim 4.12. $\pi(X) \subset \mathbb{P}^{n-1}$ is a projective variety.

Let's develop a basic tool for the proof of this claim: the **resultant** of two polynomials. Take k a field (we can suppose for now that it is algebraically closed). We have two polynomials $f(x) = \sum_{i=0}^m a_i x^i, g(x) = \sum_{j=0}^n b_j x^j$.

Question 4.13. *In terms of a_i, b_j , do f, g have a common zero?*

Let $V_d := \{\text{polynomials of degree } \leq d \text{ in } x\}$ (this is a vector space). Look at the linear map

$$\begin{aligned} V_{n-1} \times V_{m-1} &\xrightarrow{\phi} V_{n+m-1} \\ (a, b) &\mapsto af + bg. \end{aligned}$$

Now, we have a map between two vector spaces of the same dimension $(n + m)$. If f and g have a common zero, this map cannot be surjective. Conversely, if the map is not surjective, we can take an element of the kernel to find a common zero.

So, to answer the question above, the polynomials f and g have a common zero if and only if ϕ is singular, i.e. $\det \phi = 0$.

5 Lecture 5 (2/10/26)

Recall the following setup from the previous lecture: We have a point $p \in \mathbb{P}^n$ and a hyperplane $H \subset \mathbb{P}^n$ with $p \notin H$. Up to a change of coordinates, we can let $p = [0, \dots, 0, 1]$ and $H = V(Z_n)$.

Definition 5.1. Let $X \subset H$ be any variety. Then, we can form the cone

$$\overline{p, X} := \bigcup_{q \in X} \overline{pq}.$$

Remark 5.2. Then, $\overline{p, X}$ is a projective variety. Namely, taking the coordinates as above, we have that X is the zero locus of $\{F_\alpha\}$, then so is $\overline{p, X}$ (albeit in a different space).

More generally, if we have $X, Y \subset \mathbb{P}^n$ projective varieties with $X \cap Y = \emptyset$, we can form the join

$$J(X, Y) := \bigcup_{p \in X, q \in Y} \overline{pq}.$$

Proposition 5.3. $J(X, Y)$ is a projective variety.

We don't currently have the machinery to prove this proposition, but we will get there in a few lectures.

Example 5.4. Take $Y = \{p\}$ where $p \notin X$. We will prove Proposition 5.3 for this setup.

Definition 5.5. Take a point $p \in \mathbb{P}^n$, and a hyperplane $H \cong \mathbb{P}^{n-1}$ with $p \notin H$.

Let $X \subset \mathbb{P}^n$ be a projective variety and assume $p \notin X$.

We can then define the projection map

$$\begin{aligned} \pi_p : \mathbb{P}^n \setminus \{p\} &\rightarrow H \\ q &\mapsto \overline{qp} \cap H. \end{aligned}$$

If we again take the coordinates to be $p = [0, \dots, 0, 1]$ and $H = V(Z_n)$, then the projection map is quite easy to write down:

$$[Z_0, \dots, Z_n] \mapsto [Z_0, \dots, Z_{n-1}].$$

These coordinates make it fairly easy to see why it is important for our domain to be $\mathbb{P}^n \setminus \{p\}$. Otherwise, we could have a point in the output with all coordinates zero!

Claim 5.6. $\pi_p(X) \subset H \cong \mathbb{P}^{n-1}$ is again a projective variety.

5.1 Elimination Theory/Resultants

Question 5.7. Let K be an algebraically closed field. Let

$$\begin{aligned} f &= a_0 + a_1 z + \dots + a_m z_m \\ g &= b_0 + b_1 z + \dots + b_m z_m. \end{aligned}$$

We ask: do f, g have a common zero?

Answer 5.8. Let $V_d = \{\text{polynomials of degree at most } d \text{ in } z\}$. This is a vector space over K of dimension $d + 1$. Now, define the map

$$\begin{aligned}\phi : V_{n-1} \times V_{m-1} &\rightarrow V_{n+m-1} \\ (a, b) &\mapsto fa + gb\end{aligned}$$

Then, f and g have a common zero if and only if the above map is singular, that is $\det \phi = 0$.

To this end, we can choose a basis $\{1, z, \dots, z_l\}$ for V_l .

Then, ϕ is represented by the $(m+n) \times (m+n)$ matrix:

$$\begin{pmatrix} a_0 & a_1 & \dots & a_m & 0 & 0 & \dots & 0 \\ 0 & a_0 & \dots & a_{m-1} & a_m & 0 & \dots & 0 \\ \vdots & & & \vdots & & & & \vdots \\ 0 & 0 & \dots & 0 & a_0 & a_1 & \dots & a_m \\ b_0 & b_1 & b_2 & \dots & b_n & 0 & \dots & 0 \\ \vdots & & & \vdots & & & & \vdots \\ 0 & \dots & 0 & b_0 & b_1 & \dots & b_{n-1} & b_n \end{pmatrix}$$

Definition 5.9. $\det(\phi)$ is called the **resultant** of f, g , denoted $R(f, g)$.

Now replace K by $k[z_0, \dots, z_{n-1}]$, with $f, g \in k[z_0, \dots, z_n]$. This means we can write

$$\begin{aligned}f(z_n) &= a_0(z_0, \dots, z_{n-1}) + a_1(z_0, \dots, z_{n-1})z_n + \dots \\ g(z_n) &= b_0(z_0, \dots, z_{n-1}) + \dots\end{aligned}$$

We have thus formed the same map, not as a map of vector spaces, but as a map of modules over the ring $k[z_0, \dots, z_{n-1}]$. This allows us to write down the same matrix, but with the entries as polynomials in $(z_0, \dots, z_{n-1})$ rather than as scalars.

We conclude that the locus of $(z_0, \dots, z_{n-1})$ such that f, g have a common zero is the resultant with respect to z_n , denoted $R_{z_n}(f, g) \in k[z_0, \dots, z_{n-1}]$. Then, the zero locus of $R_{z_n}(f, g)$ is equal to

$$\{[z_0, \dots, z_{n-1}] : F(z_0, \dots, z_{n-1}, z_n), G(z_0, \dots, z_{n-1}, z_n) \text{ have common root in } z_n\}.$$

for all homogeneous pairs $F, G \in I(X)$.

Now, let's return to our discussion of projection maps using these new tools. Once again let $p = [0, \dots, 0, 1]$ and $H = V(Z_n)$. We have

$$\pi_p(X) = V\{R_{z_n}(f, g) : f, g \in I(X)\}.$$

So, we can conclude that $\pi_p(X)$ is a projective variety.

In a few weeks, we're going to prove the much more general statement that the join of two projective varieties is a projective variety.

We can prove a more general statement. Take $p \in \mathbb{P}^n$, hyperplane $H \subset \mathbb{P}^n$, and variety

$X \subset \mathbb{P}^n$ where $p \notin X, H$. Our claim is that

$$\overline{pX} = \bigcup_{q \in X} \overline{pq}$$

is a projective variety. This is clear from that fact that $\overline{pX}$ is exactly the cone at p over $\pi_p(X)$, which we have already shown is a variety.

The key consequence of our elimination theory:

Claim 5.10. Let Y be any variety and let $X \subset Y \times \mathbb{P}^n$ be any closed subvariety. Then, $\pi_1(X) \subset Y$ is closed.

This has lots of interesting consequences, a few of which we will cover:

- If $X \subset \mathbb{P}^n$ is a connected projective variety with f any regular function on X , then f is constant.

Proof. f defines a regular map $X \rightarrow \mathbb{A}^1 \subset \mathbb{P}^1$ whose image is a connected closed subvariety of $\mathbb{P}^1$ contained in $\mathbb{A}^1$. $\square$

This is the analog of learning there are no non-constant everywhere holomorphic functions on the Riemann sphere.

- Suppose $X \subset \mathbb{P}^n$ is a connected projective variety and $X \neq \{p\}$. Suppose $Y = V(F)$, where $F \in k[Z_0, \dots, Z_n]$ hom. polynomial. Then, $X \cap Y \neq \emptyset$.

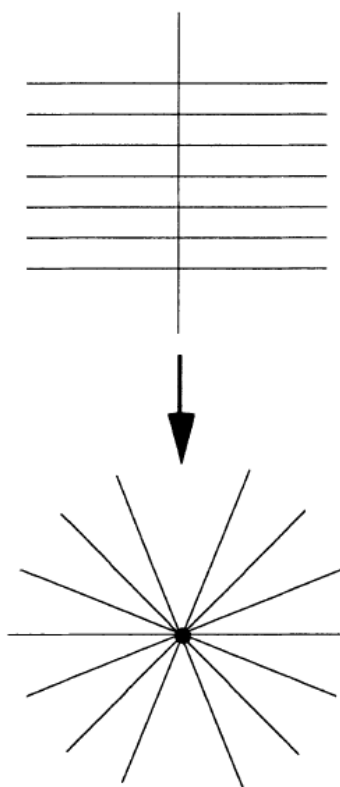
Proof. If $X \cap Y = \emptyset$, then for any G homogeneous polynomial with $\deg G = \deg F$, the ratio G/F is a regular function on F , and thus must be a constant. Since this is true for all G , X must be a point. $\square$

The statements we have made in this lecture fail in the affine setting.

Example 5.11. Consider the regular map

$$\begin{aligned}\nu : \mathbb{A}^2 &\rightarrow \mathbb{A}^2 \\ (x, y) &\mapsto (x, xy).\end{aligned}$$

The images of horizontal lines ($y = k$) correspond to lines through the origin. So, the image of ν does not contain the y -axis. More specifically, we have $\text{im } \nu = \{(x, y) : x \neq 0\} \cup \{(0, 0)\}$. This is not closed! It isn't even locally closed at the origin. This is not a quasi-projective variety.



The images of regular maps in affine space are just not well behaved. Hopefully this motivates our use of projective space.

We will now preview the very important idea of families of varieties.

Definition 5.12. A family of projective varieties parameterized by B is a closed subvariety $\mathcal{X} \subset B \times \mathbb{P}^n$ together with a projection map $\pi_1 : \mathcal{X} \rightarrow B$ for $\mathcal{X} \subset B \times \mathbb{P}^n$.

So, for each point $b \in B$, we can take the fiber to get a projective variety in $\mathbb{P}^n$. This is what we mean by the parameterization by B .

6 Lecture 6 (2/12/26)

Recall the definition of a family of varieties from the last lecture. As it stands, this construction does not seem to be particularly useful. In particular, it is not clear that these fibers have to have anything to do to one another. But, there are conditions we can impose such that these varieties vary continuously. The most important condition is one of *flatness*, which we'll discuss more later in the course. We won't bother ourselves with that right now: instead we'll look at examples.

6.1 Some Examples

Example 6.1 (The Universal Hyperplane). Take $\mathbb{P}^n = \mathbb{P}V$ for some $n + 1$ dimensional vector space V . Then, we have

$$\{\text{hyperplanes in } \mathbb{P}^n\} = \mathbb{P}V^*$$

where the correspondence is due to the fact that a hyperplane is equivalent to the zero locus of a single linear form, which is indexed by an element of $\mathbb{P}V^*$.

We claim that these form a family. Let

$$\Sigma := \{(H, p) \in \mathbb{P}V^* \times \mathbb{P}V : p \in H\} \subset \mathbb{P}V^* \times \mathbb{P}V$$

If we choose a basis $Z_0, \dots, Z_n$ for V and the dual basis $W_0, \dots, W_n$ for V^* . Then, we can write

$$\Sigma = \{([w_0, \dots, w_n], [z_0, \dots, z_n]) : \sum z_i w_i = 0\} \subset \mathbb{P}^{n*} \times \mathbb{P}^n$$

Example 6.2 (The Universal Hyperplane Section). If $X \subset \mathbb{P}^n$ is a projective variety, the hyperplane sections of X form a family. The hyperplane sections of X are exactly $\alpha^{-1}(X) \subset \mathbb{P}^{n*} \times X$.

$$\begin{array}{ccc} & \Sigma \subset \mathbb{P}^{n*} \times \mathbb{P}^n & \\ \pi \swarrow & & \searrow \alpha \\ \mathbb{P}^{n*} & & \mathbb{P}^n \supset X \end{array}$$

Example 6.3 (The family of hypersurfaces of degree d in $\mathbb{P}^n$). Let W be the vector space of homogeneous polynomials of degree d in $Z_0, \dots, Z_n$. Alternatively, we can think of this as $\text{Sym}^d V^*$, where $\mathbb{P}^n = \mathbb{P}V$.

Define

$$\Phi = \{(F, Z) \in \mathbb{P}W \times \mathbb{P}^n : F(Z) = 0\}.$$

We then have a projection map $\Phi \rightarrow \mathbb{P}W$ where the fiber is exactly the zero locus of the given polynomial.

These examples are still quite general. Let's look at a simple concrete example.

Example 6.4 (Universal family of conics). Take $d = n = 2$ from Example 6.3. Then,

$$\Phi := \{([a, b, c, d, e, f], [x, y, z]) : ax^2 + by^2 + cz^2 + dxy + exz + fyz = 0\} \subset \mathbb{P}^5 \times \mathbb{P}^2.$$

We have a projection map $\Phi \rightarrow \mathbb{P}^5$ where each fiber corresponds to the zero locus of the conic.

Now, not all of the fibers look the same. Up to projective equivalence, there are three types of quadrics. There are irreducible quadratics (rank 3), products of distinct linear terms (rank 2), and squares of linear terms (rank 1).

Professor Harris would like to get slightly “into the woods” on this last example. Consider the locus $\Gamma \subset \mathbb{P}^5$ of polynomials F such that F is the square of a linear form. Such polynomials are exactly of the form

$$(\alpha X + \beta Y + \gamma Z)^2 = \alpha^2 X^2 + 2\alpha\beta XY + \dots$$

Then Γ is just the image of the map $\mathbb{P}^2 \rightarrow \mathbb{P}^5$ which takes $[\alpha, \beta, \gamma] \mapsto [\alpha^2, 2\alpha\beta, \beta^2, 2\beta\gamma, \gamma^2, 2\alpha\gamma]$. This map is exactly the Veronese map (up to a couple extra factors of 2). So, the squares of linear forms define a closed subvariety of $\mathbb{P}^5$ isomorphic to the Veronese surface.

We can do similar work for quadrics which are the product of linear factors. This locus in $\mathbb{P}^5$ will be a projection of the Segre variety. On the open complement of this subvariety, all of the conics are irreducible “conic-looking” quadrics. This is a good example in that our fibers will often not look identical, and the different types of fibers will often correspond to closed subvarieties.

Let’s look at one more example motivated by the following question.

Question 6.5. *Do the lines in $\mathbb{P}^3$ form a family of varieties?*

Example 6.6 (A family of lines in $\mathbb{P}^3$). We want $\mathcal{L} \subset B \times \mathbb{P}^3$ with a projection onto B such that the fibers are exactly the lines in $\mathbb{P}^3$. Let $B = \mathbb{P}^3 \times \mathbb{P}^3 \setminus \Delta$, and

$$\mathcal{L} = \{(p, q, r) \in (\mathbb{P}^3 \times \mathbb{P}^3 \setminus \Delta) \times \mathbb{P}^3 : r \in \overline{pq}\}.$$

We are looking at $\{(p, q, r) \in \mathbb{P}^3 \times \mathbb{P}^3 \times \mathbb{P}^3 : p, q, r \text{ linearly dependent}\}$. This is equivalent to saying that all 3×3 minors of the matrix

$$\begin{pmatrix} a_0 & \dots & a_3 \\ b_0 & \dots & b_3 \\ c_0 & \dots & c_3 \end{pmatrix}$$

vanish. So, $\mathcal{L}$ is a variety, which is good. The issue is that every line appears many times as a fiber. For any given line, there are many pairs of points that define that line. What we would really like is to have a family where each line appears exactly once as a fiber. We can in fact find such a family. This is the grassmannian. We will look at grassmannians in depth next week or so.

6.2 A little algebra

For this, we will start with the affine case. The projective case for this is similar, just more confusing.

We start with a “simple minded question” according to Professor Harris.

Question 6.7. *Let's say $X \subset \mathbb{A}^n$ is an affine variety. What do we mean when we say that a collection of polynomials $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ "cut out" the variety X ?*

"This sounds like a stupid question, and in some sense it is." -Prof. Harris

The simplest answer is that $X = V(f_1, \dots, f_m)$. There is another possible meaning. We could also write that the ideal $I(X) = (f_1, \dots, f_m)$. These two notions are not the same. The standard example is for $n = 1$, $X = \{0\} \subset \mathbb{A}^1$. The polynomial x^2 has zero locus X , but the ideal $I(X) = (x)$ is not generated by x^2 . So, we have some correspondence between varieties and ideals, but the correspondence is not one to one.

Observe: not every ideal $I \subset k[x_1, \dots, x_n]$ is the ideal of a variety X . Again, the easiest example is that for any variety $X \subset \mathbb{A}^n$, we have that $I(X)$ is a radical ideal.

Definition 6.8. We say that an ideal I is **radical** if $f^n \in I$ implies $f \in I$. Equivalently, $k[x_1, \dots, x_n]/I$ has no nilpotent elements (the set of nilpotent elements is called the **nilradical** - it is the intersection of all prime ideals of a ring).

So, we have a correspondence

$$\{\text{Varieties } X \subset \mathbb{A}^n\} \rightleftarrows \{\text{ideals } I \subset k[x_1, \dots, x_n]\}$$

where the forward direction is given by $I(X)$ and the reverse is given by $Z(I)$. Note this is not a bijection.

What we do have is the following major theorem (which we will prove next week).

Theorem 6.9 ((one version of the) Nullstellensatz). $I(V(I)) = \text{rad}(I) := \{f \in k[x_1, \dots, x_n] : f^r \in I \text{ for some } r\}$

6.3 An aside motivating schemes

"It's really pretty simple" - Prof. Harris.

For centuries, people thought that the way to go was to restrict to only considering radical ideals. In this setting, we do have a bijection. Then, Grothendieck came along and told everyone that they were wrong. He expanded the notion of varieties instead of restricting the set of ideals. These are schemes.

Let's talk about a concrete situation where this arises. Let's say in the plane we have a line and a conic. The question is: at how many points do they intersect?

By moving to $\mathbb{C}$, we have that the line and conic intersect at two points. The issue is that this counts multiplicity. So, the line could be tangent to the conic. Let's look at the ideal of the intersection. If the two intersection points are distinct, the ideal of the intersection is exactly generated by the ideals of the line and the conic. The issue is that when we look at the tangent case, this is not true! The ideal of the line and the conic generate a subideal. This confusion can all be fixed with schemes. We will not touch that (take Math 232A if you're interested).

6.4 Back to course material

Looking forward, we will prove Nullstellensatz. In order to do that, we will need some language and some results from commutative algebra.

Definition 6.10 (Irreducible varieties). We say a variety X is **irreducible** if for all pairs $Z, W \subset X$ closed, $X = Z \cup W \implies X = Z$ or $X = W$.

Proposition 6.11. *If X is affine, then X is irreducible if and only if $I(X)$ is prime.*

Proof. To see irreducibility implies $I(X)$ prime, suppose $f = gh \in I(X)$, and neither $g, h \in I(X)$. Then note that $(Z(g) \cap X) \cup (Z(h) \cap X) = X$, but neither closed subvarieties are equal to X . To see the other direction, suppose $I(X)$ is prime. Consider $Z \cup W = X$ a pair of varieties. If $Z \neq X, \exists h \in I(Z) \setminus I(X)$. We can choose similar g if $W \neq X$. But then $gh \in I(X)$, contradiction. $\square$

Theorem 6.12 (Commutative algebra fact). *If $I \subset k[x_1, \dots, x_n]$ and I is radical. Then, I is uniquely expressible as the intersection of prime ideals.*

The more general statement (which you will find in a “grown-up algebra textbook”) drops the condition on radical ideals and involves primary ideals.

Corollary 6.13. *Every variety can be uniquely written as the union of irreducible varieties.*

7 Lecture 7 (2/17/26)

Today, we will talk about two theorems from commutative algebra and their consequences, including Grassmannians.

7.1 Radical Ideals and the Nullstellensatz

Definition 7.1. The **radical** of an ideal I is given by

$$r(I) = \{f \in R \mid f^n \in I \text{ for some } n > 0\}$$

This is again an ideal. We say I is radical if $r(I) = I$. This is equivalent to saying that R/I has no nilpotents, or is nilpotent-free.

Remark 7.2. Ideals of varieties are always radical; if some power of a polynomial vanishes on the variety, then so too must the polynomial.

This gives us a two-way correspondence (in the case of affine varieties and polynomial rings):

$$\{\text{varieties } X \subset \mathbb{A}^n\} \longleftrightarrow \{\text{ideals } I \subset k[z_1, \dots, z_n]\}$$

where the arrows are given by taking ideals and zero loci respectively.

We can understand one direction of this correspondence quite easily. By definition, $V(I(X)) = X$. For the other direction, we have the Nullstellensatz (or at least one version of it).

Theorem 7.3. For any ideal $\mathfrak{A} \in k[z_1, \dots, z_n]$, we have

$$I(V(\mathfrak{A})) = r(\mathfrak{A}).$$

We can deduce this theorem from an a priori weaker statement.

Theorem 7.4. If $V((A)) = \emptyset$, then $\mathfrak{A} = (1) = k[z_1, \dots, z_n]$.

This statement follows from the following seemingly even weaker statement.

Theorem 7.5. Every maximal ideal in $k[z_1, \dots, z_n]$ is of the form $(z_1 - a_1, z_2 - a_2, \dots, z_n - a_n)$.

To see how these theorems fit together to prove each other, see Harris Lecture 5.

Take polynomials $f_1, \dots, f_m \in k[z_1, \dots, z_m]$. Suppose $X \subset \mathbb{A}^n$ is an affine variety.

Remark 7.6. Say $f_1, \dots, f_m$ “cut out” the variety X *set-theoretically* if $V(f_1, \dots, f_m) = X$.

Alternatively, we say $f_1, \dots, f_m$ “cut out” the variety X *scheme-theoretically* if $(f_1, \dots, f_m) = I(X)$.

This comes back to the discussion from last week, where the Nullstellensatz proposes making the correspondence a bijection by restricting the class of ideals to radical ideals, whereas Grothendieck suggest enlarging the class of varieties to schemes.

7.2 Ideals of Projective Varieties

Let $S = k[Z_0, \dots, Z_n]$.

Definition 7.7. We have that S is a **graded ring**: if we let

$$S_d = \{\text{homogeneous degree } d \text{ polynomials}\} \subset S,$$

we can then see that

$$S = \bigoplus_{d=0}^{\infty} S_d.$$

Abstractly, if V is a $n + 1$ dimensional vector space over k . Then, we define

$$S := \bigoplus_{d=0}^{\infty} \text{Sym}^d V^*$$

We say that an ideal $I \subset S$ is a graded or homogeneous ideal if $I = \bigoplus (I \cap S_d)$.

From this, another issue arises that we will discuss briefly. Suppose we have a homogeneous graded ideal. We write the d^{th} graded component $I \cap S_d$ as I_d . The idea is that we can truncate such an ideal I_d as follows: for any ℓ , we introduce the **truncated** ideal

$$I' = \bigoplus_{d=\ell}^{\infty} I_d$$

given by taking a homogeneous graded ideal and “chopping off” all pieces $d < \ell$. Note that this will not change the common zero locus (think about why). To go the other direction, we can introduce the following

Definition 7.8 (Saturated Ideals). The **saturation** of a graded ideal is

$$\text{sat}(I) := \{f \mid f \cdot (Z_0, \dots, Z_n)^m \subset I \text{ for some } m > 0\}$$

where $(Z_0, \dots, Z_n)$ is the ideal generated by $(Z_0, \dots, Z_n)$. We say that I is **saturated** if $I = \text{sat}(I)$.

This serves to reverse the truncation, since if we truncate an ideal and then take the saturation, we get back all the graded pieces.

Now, given $X \subset \mathbb{P}^n$ and $I \subset k[Z_0, \dots, Z_n]$ graded ideal, we can say three things:

- We can say that I cuts out X set-theoretically if $V(I) = X$.
- We say I cuts out X scheme-theoretically if $\text{sat}(I) = I(X)$.
- The strongest of these: $I = I(X)$ (sub-case of the second item).

Example 7.9. By way of example, in $\mathbb{P}^2$, we have $V(X, Y) = \{[0, 0, 1]\}$. By contrast, look at the ideal of $V(X^2, Y)$. This ideal is saturated, but not radical. Note in particular that $X \notin \text{sat}(X^2, Y)$, because $XZ^m \in (X^2, Y) \forall m > 0$, but $XZ^m \notin (X^2, Y)$.

7.3 Irreducibility

We now have, as promised, our second commutative algebra theorem.

Theorem 7.10. *Any radical ideal $I \subset k[z_1, \dots, z_n]$ is uniquely expressible as an irredundant intersection of prime ideals. This means that we write*

$$I = \bigcap \mathfrak{p}_\alpha$$

such that $\mathfrak{p}_\alpha \not\subset \mathfrak{p}_\beta$ for all $\alpha \neq \beta$.

If you're curious about more such commutative algebra, take a look at Eisenbud's Commutative Algebra or Atiyah-Macdonald.

Moving on to the geometric significance of this, we say:

Definition 7.11. A variety X is **irreducible** if for all $Y, Z \subset X$ closed,

$$Y \cup Z = X \implies Y = X \text{ or } Z = X.$$

This is equivalent to the statement that $I(X)$ is prime.

This characterization allows us to restate the above Theorem 7.10 in geometric terms:

Theorem 7.12. *Any affine variety $X \subset \mathbb{A}^n$ is uniquely expressible as an irredundant union of irreducible varieties.*

One serious warning. The world of AG is split into some people, like Hartshorne, who believe that the word variety means irreducible variety. These people call a general variety an algebraic set. Professor Harris finds such language ungainly, and thus we will say that a variety is the zero locus of any collection of polynomials.

7.4 General Objects

We now talk a little bit about some language. Recall our discussion of families of varieties parameterized by a variety from last week.

Definition 7.13. A **family of projective varieties parameterized by B** is a closed subvariety $\mathfrak{X} \subset B \times \mathbb{P}^n$ together with a projection map $\pi_1 : \Sigma \rightarrow B$ for $\Sigma \subset B \times \mathbb{P}^n$. We also define the **members of the family** to be $\{\pi^{-1}(b) \mid b \in B\}$.

Now, suppose we're considering a particular property, and want to ask which members of the family possess that property.

Definition 7.14 (General Object). We say a **general member** of this family has **property P** if the set $\{b \mid X_b \text{ has property } P\}$ contains a dense open subset of B .

Remark 7.15. In irreducible varieties, all open sets are dense (this is on the HW). So if B is irreducible, this is equivalent to saying that the set of points *not* possessing this property is contained in a proper closed subvariety.

Example 7.16. We claim that a general triple of points $(p, q, r) \in \mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^2$ are not collinear. This can be written as

$$\begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix}$$

where our three points are collinear iff the determinant vanishes. The determinant will be homogeneous with respect to each row, which means that the locus of collinear triples is a proper closed subvariety of the product. (This determinant-vanishing-trick is something we've seen a couple of times now, and is frequently useful in showing that the set of points satisfying a given property is closed).

We see that talking about general objects implies the existence of a family of such objects.

Example 7.17. We claim that a general conic curve $C \subset \mathbb{P}^2$ is irreducible. We have a family of conics

$$V(aX^2 + bY^2 + cZ^2 + dXY + eXZ + fZ^2) = \mathcal{C} \subset \mathbb{P}^5 \times \mathbb{P}^2$$

with points in $\mathbb{P}^5 \times \mathbb{P}^2$ given by (C, x) such that $x \in V(C)$.

8 Lecture 8 (2/19/26)

Recall the following claim.

Claim 8.1. If $X \subset \mathbb{P}^n$ is a projective variety, we can write X as the zero locus of homogeneous polynomials all of the same degree.

Proof. By definition, X is the zero locus of some finite collection $V(F_1, \dots, F_m)$. For any homogeneous ideal $I \subset K[Z_0, \dots, Z_n]$ we can define the truncation

$$I_d := \bigoplus_{e \geq d} I_e.$$

Observe that the zero locus of the truncation is exactly equivalent to the zero locus of the original ideal, as the product of generators of I times all suitable degree monomials will be in the truncation. Truncating to the maximum degree of the F_α gives the result. $\square$

If we have a closed subvariety of $\mathbb{P}^n \times \mathbb{P}^m$, this gives a closed subvariety in $\mathbb{P}^{mn+m+n}$ where homogeneous polynomials pull back to bihomogeneous polynomials in $\mathbb{P}^n \times \mathbb{P}^m$. If we want to have the bidegrees be balanced, we can perform the same trick as in the claim above.

8.1 Grassmannians

Definition 8.2. Our basic objects are **grassmannians**:

$$G(k, n) := \{k\text{-dim vector subspaces of } K^n\}.$$

For V any n dimensional vector space over K , we also write $G(k, V)$.

When we projectify, we say

$$\mathbb{G}(k-1, n-1) := \{(k-1)\text{-planes in } \mathbb{P}^{n-1}\}.$$

Similarly, we define $\mathbb{G}(k-1, \mathbb{P}V) := \{(k-1)\text{-planes in } \mathbb{P}V\}$.

Let's consider this topologically. We can think of $G_{\mathbb{C}}(k, n)$ as a topological space/manifold. We defined

$$\mathbb{CP}^{n-1} = \mathbb{C}^n \setminus \{0\} / \mathbb{C}^*.$$

We can think of the grassmannian similarly. For $G(k-1, n-1)$, we start with $k \times n$ matrices of complex numbers of rank k , and then mod out by $\text{GL}_k \mathbb{C}$, as permutations of the k space doesn't change the span of our matrix. Put together, we have

$$G(k-1, n-1) = \{k \times n \text{ matrices of rank } k\} / \text{GL}_k \mathbb{C}.$$

We can cover the grassmannian with open subsets $\{k \times n \text{ matrices with a specified minor nonzero}\}$. For example, if we are in the open set where the first minor is non-zero, we can change co-

ordinates and write out a matrix in $G(k-1, n-1)$ uniquely in the form

$$\left(\begin{array}{cc|ccc} 1 & & 0 & z_{1,k+1} & \cdots & z_{1,n} \\ & \ddots & & \vdots & \ddots & \vdots \\ 0 & & 1 & z_{k,k+1} & \cdots & z_{k,n} \end{array} \right).$$

Remark 8.3. As a fun factoid to think about, $\mathbb{G}_{\mathbb{C}}(k, n)$ has the structure of a CW complex with $\binom{n}{k}$, all of even dimension.

8.2 Exterior Algebras

Now, we will take a diversion into some algebra.

Fix some n dimensional vector space V over K .

Definition 8.4. We define the exterior product

$$\Lambda^k V := \{\text{skew-symmetric elements of } \underbrace{V \otimes \cdots \otimes V}_{k \text{ times}}\}.$$

We then define the **exterior algebra**

$$\Lambda^* V := \bigoplus_k \Lambda^k V = \bigoplus_{i=0}^n \Lambda^i V.$$

Observe that $\Lambda^n V \cong K$. Note that we get a nondegenerate bilinear map $\Lambda^k V \times \Lambda^{n-k} V \rightarrow \Lambda^n V \cong K$.

Lemma 8.5. Take $\omega \in \Lambda^k V, v \in V$ with $v \neq 0$. Then, we can write $\omega = v \wedge \phi$ for some $\phi \in \Lambda^{k-1} V$. if and only if $\omega \wedge v = 0$.

Now, we have a map

$$G(k, V) \rightarrow \mathbb{P}\Lambda^k V$$

defined by $\langle v_1, \dots, v_k \rangle \mapsto [v_1 \wedge \cdots \wedge v_k] \in \mathbb{P}\Lambda^k V$. Note that this map is well defined, as a different choice of basis on the left changes the right side by a factor of the determinant. Since we are in projective space, this scalar factor does not matter. We also want to show that this map is an embedding. To see that this is injective, observe that if $\omega = v_1 \wedge \cdots \wedge v_k$. Then, given $[\omega]$, we can recover $\Lambda \subset V$ as $\Lambda = \{v \in V : v \wedge \omega = 0\}$. So, this map is injective.

Claim 8.6. The element $\omega \in \Lambda^k V$ is in the image of $G(k, n)$ (so $\omega = v_1 \wedge \cdots \wedge v_k$) if and only if the rank of the map $V \xrightarrow{\wedge \omega} \Lambda^{k+1} V$ is at most $n - k$.

Look at the map $V \times \Lambda^k V \mapsto \Lambda^{k+1} V$ such that $(v, \omega) \mapsto v \wedge \omega \in \Lambda^{k+1} V$. This map generates a linear map

$$\varphi : \Lambda^k V \rightarrow \text{Hom}(V, \Lambda^{k+1} V).$$

We can then define $G(k, V) := \{\omega \in \Lambda^k V : \text{rank } \varphi(\omega) \leq n - k\}$. This is enough to show that the g=Grassmannian is a projective variety, but these polynomials do not generate the ideal. The ideal is in fact generated by quadrics!

Now, let's consider some different constructions. Recall that the join of two projective varieties X and Y is the union of all lines that intersect both varieties. We will be able to use Grassmannians to show that the join of any varieties is again a variety.

Claim 8.7. The set

$$\Sigma := \{(\Lambda, p) \in \mathbb{G}(k, n) \times \mathbb{P}^n : p \in \Lambda\}$$

form a family parameterized by the Grassmannian $\mathbb{G}(k, n)$.

9 Lecture 9 (2/24/26)

Some logistical notes: the midterm will be on March 10th, and will cover chapters 1 – 7 of the textbook.

9.1 Brief Recap

Now, onto the math. Let's recall some of the definitions and such from last week about the Grassmannian $G(k, v) = \{\wedge^k \subset V\}$.

To start, we note that the grassmannian has the structure of a projective variety via the inclusion $G(k, v) \hookrightarrow \mathbb{P} \wedge^k V$ defined by sending

$$\langle v_1, \dots, v_k \rangle \mapsto [v_1 \wedge \dots \wedge v_k]$$

. Note that this map is injective. This can be seen by taking the wedge with arbitrary vectors, due to the fact that $\omega \wedge v = 0$ if and only if $\omega = v \wedge \phi$.

To see this in more detail, note that we have a natural linear map $V \otimes \wedge^k V \rightarrow \wedge^{k+1} V$. We take the transpose of that map and get a map $\varphi : \wedge^k V \rightarrow \text{Hom}(V, \wedge^{k+1} V)$. Then, $G(k, v) = \{\omega \in \wedge^k V : \text{rank} \varphi(\omega) \leq n - k\}$. This condition on the rank is a homogeneous polynomial condition on $\mathbb{P} \wedge^k V$ (given by the minors of the corresponding matrices). As noted in the previous lecture, these polynomials do not generate the vanishing ideal. In fact, the ideal of the Grassmannian is generated by homogeneous quadratic polynomials, called **Plücker relations**. We're not going to discuss these in detail in class.

9.2 The Grassmannian as a parameter space

Look at $\mathbb{G}(k, V) = \{\text{planes } \Lambda \cong \mathbb{P}^k \subset \mathbb{P}V \cong \mathbb{P}^n\}$. The basic claim is that the Grassmannian is a parameter space for linear subspaces. That means that the set of k -planes in a fixed projective space form a family over the Grassmannian.

Claim 9.1. $\Sigma = \{(\Lambda, p) : p \in \Lambda\} \subset \mathbb{G}(k, V) \times \mathbb{P}V$ is a closed projective subvariety where the fibers over a point in $\mathbb{G}(k, v)$ are exactly the k -planes.

Proof: (Just a little multilinear algebra).

$$\mathbb{G}(k, V) \times \mathbb{P}V \supset \Sigma = (\mathbb{G}(k, V) \times \mathbb{P}V) \cap \{(w, v) : w \wedge v = 0\} \subset \mathbb{P} \wedge^{k+1} V \times \mathbb{P}V$$

□

The first example of a Grassmannian that is not already a projective space is $G(2, 4)$ (2-dimensional subspaces of K^4) or equivalently $\mathbb{G}(1, 3)$ (lines in $\mathbb{P}^3$).

9.3 Some consequences

Example 9.2. If $\Phi \subset \mathbb{G}(k, v)$ is a projective subvariety, then we can take the union $\Psi = \bigcup_{\Lambda \in \Phi} \Lambda \subset \mathbb{P}V$. We claim this is a projective subvariety.

Proof. Consider the projections:

$$\begin{array}{ccc}
& \Sigma \subset \mathbb{G}(k, V) \times \mathbb{P}V & \\
\swarrow \pi_1 & & \searrow \pi_2 \\
\mathbb{G}(k, V) & & \mathbb{P}V
\end{array}$$

Then $\Psi = \pi_2(\pi_1^{-1}(\Phi))$.

□

Example 9.3 (Incident k -planes). Suppose we start with a subvariety $X \subset \mathbb{P}V$. We can associate to that a subset of the Grassmannian by looking at the k -planes which meet that variety. In particular, define the **locus of incident k -planes** $\mathcal{C}_k(X) := \{\Lambda \in \mathbb{G}(k, V) : \Lambda \cap X \neq \emptyset\} \subset \mathbb{G}(k, V)$. We claim $\mathcal{C}_k(X)$ is a closed projective subvariety of $\mathbb{G}(k, V)$.

Proof. $\mathcal{C}_k(X) := \pi_1(\pi_2^{-1}(X))$ (see diagram in previous example).

□

Example 9.4 (The Join of two varieties). Suppose $X, Y \subset \mathbb{P}V$ closed subvarieties, and assume $X \cap Y = \emptyset$. We claim that

$$\bigcup_{p \in X, q \in Y} \overline{pq} = \mathcal{J}(X, Y) \subset \mathbb{P}V$$

is a projective variety.

Proof. In terms of the diagram above, we have

$$\mathcal{J}(X, Y) = \pi_2(\pi_1^{-1}(\mathcal{C}_1(X) \cap \mathcal{C}_1(Y))).$$

□

We now consider a question that we'll investigate going forward.

Question 9.5. *What if X, Y are not disjoint? Could we extend the notion of a join to such varieties?*

We go to the opposite extreme. Take $X = Y$. Consider the union of lines $\overline{pq}$ where $p, q \in X, p \neq q$, and then take the closure. This gives what is known as the **secant variety**.

Up until now, we have been talking about incidence correspondences: families defined by intersections. We can look at the other extreme: families about containment.

Claim 9.6. Given variety $X \subset \mathbb{P}V$, look at

$$F_k(X) := \{\Lambda \in \mathbb{G}(k, V) : \Lambda \subset X\}.$$

Then, $F_k(X)$ is a projective subvariety of $\mathbb{G}(k, V)$.

Definition 9.7 (Flag manifolds). A (two-step)-**Flag manifold** is the family:

$$F(\ell, k, n) := \{(\Gamma, \Lambda) \in \mathbb{G}(\ell, n) \times \mathbb{G}(k, n) : \Lambda \subset \Gamma\}$$

The claim, as always, is that $F(\ell, k, n)$ is a subvariety of $\mathbb{G}(\ell, n) \times \mathbb{G}(k, n)$.

Now, why stop at two? We can take longer nested families of k planes.

Definition 9.8. Given any sequence $0 \leq a_1 < a_2 < \cdots < a_k < n$, we define the **Flag manifold**

$$F(a_1, \dots, a_k, n) = \{(\Lambda_1, \dots, \Lambda_k) \in \Pi\mathbb{G}(a_i, n) : \Lambda_1 \subset \Lambda_2 \subset \cdots \subset \Lambda_k\}.$$

In an unsurprising twist, given the lecture so far, the general Flag manifold is a subvariety of its respective product of Grassmannians. The most extreme cases here is the **Full flag manifold**

$$F(0, 1, \dots, n-1, n) := \{(\Lambda_0, \Lambda_1, \dots, \Lambda_{n-1}) \in \mathbb{G}(0, V) \times \cdots \times \mathbb{G}(n-1, V) : \Lambda_0 \subset \cdots \subset \Lambda_{n-1}\}.$$

10 Lecture 10: 2/26/2026

10.1 Wrapping Up Grassmannians

Recall the diagram:

$$\begin{array}{ccc} & \Sigma = \{(\Lambda, p) : p \in \Lambda\} \subset \mathbb{G}(k, V) \times \mathbb{P}V & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ \mathbb{G}(k, V) & & \mathbb{P}V \end{array}$$

This will be very important for today's lecture.

We say that Σ is the “universal family of k -planes” for the following reason;

Remark 10.1. For all varieties B and $Z \subset B \times \mathbb{P}V$ such that the fibers of Z/B are k -planes in $\mathbb{P}V$, we have that there exists a map $\phi : B \rightarrow \mathbb{G}(k, V)$ such that Z is the fiber product of B with Σ via ϕ .

Recall the consequences of this diagram introduced in last lecture (Section 9.3). We add to these another example.

Example 10.2 (Joins of corresponding points). Let $X, Y \subset \mathbb{P}V$ be disjoint subvarieties. Suppose $f : X \rightarrow Y$ is a regular map. Then define

$$\Gamma := \bigcup_{p \in X} \overline{p, f(p)} \subset \mathbb{P}V$$

We claim that Γ is a subvariety.

Proof. Instead of using the same approach as in the previous three examples, we will introduce $\Phi \subset \mathbb{P}V \times \mathbb{P}V \times \mathbb{P}V$ where

$$\Phi := \{(p, q, r) : p, q, r \text{ lie on some line in } \mathbb{P}V\} \subset \mathbb{P}V^3$$

We ask, is this a variety? It should be clear, at this point, that the fact we've bothered defining it means that the answer is yes.

To see this, we can take $p = [p_0, \dots, p_n], q = [q_0, \dots, q_n], r = [r_0, \dots, r_n]$.

The statement that they lie on some line is equivalent to saying that the matrix with these points as the rows

$$\begin{bmatrix} p_0 & \dots & p_n \\ q_0 & \dots & q_n \\ r_0 & \dots & r_n \end{bmatrix}$$

has rank at most 2, i.e. that the 3×3 minors vanish. These minors are each tri-linear regular functions of the points p_i, q_i, r_i and thus their common zero locus is a subvariety. To conclude, we have the following diagram:

$$\begin{array}{ccc} & \Phi & \\ \swarrow \pi_1 & & \searrow \pi_{23} \\ \mathbb{P}V & & \mathbb{P}V \times \mathbb{P}V \supset X \times Y \supset \Gamma_f \end{array}$$

Now, we just observe that

$$\Gamma = \pi_1(\pi_{23}^{-1}(\Gamma_f))$$

to conclude that Γ is a variety. $\square$

A note on last lecture. Last time, we introduced the notion of "flag manifolds." Someone asked, why do we call them manifolds? The answer is that, over the complex numbers, Grassmannians are manifolds. To see this, we will describe the covering of the Grassmannian by affine open subsets. This is analogous to how we describe $\mathbb{CP}^n$ via an open covering of subsets of $\mathbb{C}^n$.

10.2 Covering of $\mathbb{G}(k, n)$ by affine spaces

We can start with the case of $\mathbb{P}^n = \mathbb{P}V$ where V is $(n+1)$ dimensional vector space. Now if $W \subset V$ is a linear subspace of codimension 1 (a hyperplane in $\mathbb{P}V$), then we claim that the locus $\mathbb{P}V \setminus \mathbb{P}W \cong \mathbb{A}^n$.

$$\mathbb{P}^n = \{z_0, \dots, z_n \neq (0, \dots, 0)\} / (C^* = GL_1\mathbb{C})$$

Take $\mathbb{P}^n \setminus \mathbb{P}W \subset \mathbb{C}^{n+1}$ defined by $z_0 \neq 0$. Every such point has a unique representative of the form $(1, a_1, \dots, a_n)$. Prof. Harris remarks that this may seem "overly long and unnecessary." The point is we can do something similar for Grassmannians.

If we now look at $\mathbb{G}(k, V)$. Now, imagine that we choose any vector subspace W of dimension $n-k$ in V . Define $U := \{k\text{-planes } \Lambda \subset V : \Lambda \cap W = \{0\}\}$. Now, $U \cong \mathbb{A}^{(k+1)(n-k)}$. The proof is shown in coordinates. Choose coordinates such that $W = \{z : z_0, \dots, z_k = 0\}$.

Λ = row space of $(k+1) \times (n+1)$ matrix

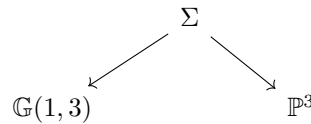
Then any Λ in U is uniquely representable by a matrix of the form

$$\left(\begin{array}{cc|ccc} 1 & & 0 & z_{1,k+1} & \dots & z_{1,n} \\ & \ddots & & \vdots & \ddots & \vdots \\ 0 & & 1 & z_{k,k+1} & \dots & z_{k,n} \end{array} \right).$$

Thus, we have an isomorphism $U \cong \mathbb{A}^{(k+1)(n-k)}$.

Question 10.3 (Simultaneous parametrization). *Every line in $\mathbb{P}^3$ is isomorphic to $\mathbb{P}^1$. Now, can we do that simultaneously for all lines?*

More precisely, we have $\Sigma := \{(\ell, p) \in \mathbb{G}(1, 3) \times \mathbb{P}^3 : p \in \ell\}$ with the following projection maps:



Our question is thus: Is $\Sigma \cong \mathbb{G}(1, 3) \times \mathbb{P}^1$?

Spoiler alert: the answer is no. In order to answer this question, we would need a lot of machinery. However, if we restrict to an open subset $U_{\ell_0} := \{\ell \subset \mathbb{P}^3 : \ell \cap \ell_0 = \emptyset\}$, then the

answer is yes!

11 Lecture 11: 3/3/26

Welcome to March! Let's wrap up our discussion of Grassmannians. We want to show one last subset is a variety.

Claim 11.1. Let X be a projective variety and let $F_k(X) := \{\Lambda \in \mathbb{G}(k, n) : \Lambda \subset X\}$. We claim that $F_k(X)$ is a closed subvariety.

Proof. To start, recall that we can cover the Grassmannian by affine open subsets, as seen in Section 10.2. $\square$

Question 11.2 (Simultaneous parameterization). Consider $\Sigma = \{(\Lambda, p) \in \mathbb{G}(k, n) \times \mathbb{P}^n : p \in \Lambda\}$ with the projection

$$\begin{array}{ccc} \Sigma = \{(\Lambda, p) \in \mathbb{G}(k, n) \times \mathbb{P}^n : p \in \Lambda\} & & \\ \swarrow & & \searrow \\ \mathbb{G}(k, n) & & \mathbb{P}^n \end{array}$$

Can we identify all fibers of π_1 simultaneously with $\mathbb{P}^k$? Alternatively, is $\Sigma \cong \mathbb{G}(k, n) \times \mathbb{P}^k$

Claim 11.3. The answer to this question is **no, but ...** In particular, if we restrict to an open subset of the form U_0 , then we can.

Consider just the case $k = 1$. Then we have

$$\begin{aligned} U_0 &= \{2 \times (n+1) \text{ matrices with first } 2 \times 2 \text{ minors} \neq 0\} \\ &= \begin{pmatrix} 1 & 0 & a_{12} & \dots & a_{1n} \\ 0 & 1 & a_{22} & \dots & a_{2n} \end{pmatrix} \end{aligned}$$

So, any line in U_0 can be given parametrically by a map $\mathbb{P}^1 \rightarrow \mathbb{P}^n$ sending

$$[s, t] \mapsto [s, t, a_{12}s + a_{22}t, \dots, a_{1n}s + a_{2n}t].$$

The point is: if $F(Z_0, \dots, Z_n)$ is any homogeneous polynomial, then we can take

$$F(s, t, a_{12}s + a_{22}t, \dots, a_{1n}s + a_{2n}t) = c_0(a)s^d + c_1(a)s^{d-1}t + \dots, c_d(a)t^d$$

, and get a polynomial in s, t . The conclusion is that $F_1(X) \cap U_0 = V(a_0, \dots, a_d)$, so there is a closed subvariety of $U_0 \subset \mathbb{G}(1, n)$.

This is the end of what we have to say about Grassmannians for the time being. They will come back.

11.1 Rational Maps

We begin with a warning: Rational maps are not in fact maps, despite the names. Many of the things you like to do with maps are not true here. You cannot compose rational maps. This is not a category.

We can talk about rational maps for all varieties, but we will start by talking only about irreducible varieties, as this will make our lives simpler. So, for the next little bit, assume the varieties are all irreducible.

Definition 11.4. Let X be an irreducible affine variety. Then, $I(X)$ is prime and $A(X) = k[z_1, \dots, z_n]/I(X)$ is an integral domain. So, we can define $K(X)$ to be the quotient field of $A(X)$ (i.e. $K(X) = \{\frac{f}{g} | f, g \in A(X), g \neq 0\}$). If $X \subset \mathbb{P}^n$, we have $S(X) := k[Z_0, \dots, Z_n]/I(X)$ is a graded ring, as $I(X)$ is a homogeneous ideal. We can take the degree 0 graded piece of $S(X)$ and take its quotient field. In symbols, this is

$$\left\{ \frac{F}{G} : \deg F = \deg G, G \not\equiv 0 \text{ on } X \right\}$$

Now, with this, we can define rational maps.

Definition 11.5. A **rational map** $f : X \rightarrow \mathbb{A}^n$ is given by an n -tuple of rational functions.

Remark 11.6. If $X \subset \mathbb{P}^m$, we can also write the map $f : X \rightarrow \mathbb{P}^n$ as

$$\varphi : [Z_0, \dots, Z_m] \mapsto [H_0(Z), \dots, H_n(Z)]$$

where the H_i are homogeneous of the same degree. If we compare this with the definition of a regular map, we see that we have dropped the restriction that the H_i do not simultaneously vanish. So, we see that the output is not necessarily a point of projective space.

Notice that the points where the map is not well-defined form a closed subset. So, we can take the following definition. This is probably the best one.

Definition 11.7. A **rational map** $f : X \rightarrow Y$ is an equivalence class of pairs (U, f_u) where U is open and $f_u : U \rightarrow Y$ is regular under the equivalence relation $(U, f_u) \sim (V, f_v)$ if $f_u|_{U \cap V} = f_v|_{U \cap V}$. Note that it is important that the varieties are irreducible so that the open sets are dense.

See the CA algebra review notes for more on these definition, which are not the most wieldy. For now, let's do an example.

Example 11.8. Consider the map $\mathbb{A}^2 \dashrightarrow \mathbb{P}^1$ where $(x, y) \mapsto [x, y]$. This map is regular on $\mathbb{A}^2 \setminus \{0, 0\}$ and undefined at the origin. Moreover, it is impossible to continuously extend the map.

Remark 11.9. We will use $\dashrightarrow$ to denote rational maps, as there are not morphisms.

Let's now introduce some associated language.

Definition 11.10. Let $\varphi : X \dashrightarrow Y$ be a rational map. Take open $U \subset X$ such that φ is regular on U , then we can form the graph

$$\Gamma_{\varphi, U} \{ (u, \varphi(u)) \} \subset U \times Y \subset X \times Y.$$

Define the graph $\Gamma_{\varphi} \subset X \times Y$ to be the closure of $\Gamma_{\varphi, U}$ for any open $U \subset X$.

We then have the following diagram.

$$\begin{array}{ccc} & \Gamma_\varphi \subset X \times Y & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ X & & Y \end{array}$$

Definition 11.11. We can define the **preimage** $\varphi^{-1}(Z) := \pi_1(\pi_2^{-1}(Z))$ and the image $\varphi(W) := \pi_2(\pi_1^{-1}(W))$

As a secondary word of warning, images and preimages do not behave as you would want.

Definition 11.12. We say X, Y are **birational** or **birationally isomorphic** if there exist rational maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $f \circ g$ and $g \circ f$ are the identity wherever they are well defined.

Alternatively, this means $K(X) \cong K(Y)$

This gives us a second, much coarser definition of isomorphism than our previous notion. With this definition, we have two main questions:

- Question 11.13.**
1. *Can we classify all varieties up to birational isomorphism?*
 2. *In a given birational isomorphism class, can we pick a “best” representative of the class? This is the minimal model program.*

An exercise to think about:

Exercise 11.14. Show, by example, that $\mathbb{P}^2$ and $\mathbb{P}_1 \times \mathbb{P}_1$ are birationally isomorphic.

12 Lecture 12: 3/5/26

Let $X \subset \mathbb{A}^n$ be irreducible. Then $A(X) := k[z_1, \dots, z_n]/I(X)$ is an integral domain. Recall from last time that $K(X)$ is the quotient field of $A(X)$. Note that $h = \frac{f}{g} \in K(X)$ need not be well-defined as a function $X \rightarrow k$ even if we allow ∞ as a value.

Example 12.1. Let $X = \mathbb{A}_{x,y}^2$ and $f(x, y) = \frac{y}{x}$. This function is not well defined everywhere, even if we allow the function to take a value of infinity! At the origin, we can get any value when we take the limit of a path towards the origin.

Let $\emptyset \neq U \subset X$ be open, nonempty. Then we define $K(U) := K(X)$. A rational function on X is the same thing as a regular function on a nonempty open $U \subset X$.

Recall the definition of $S(X)$ from the previous lecture (Def. 11.4).

Let's consider the following continuation of Example 12.1:

Example 12.2. Let $X = \mathbb{P}^2$ and consider the map $\pi : X \dashrightarrow \mathbb{P}^1$ defined by $[X, Y, Z] \mapsto [X, Y]$. This is exactly the projective equivalent of the previous exercise. This map is well defined away from $[0, 0, 1]$ and cannot be extended.

Recall Def 11.7 and Def 11.10 of rational maps and their graphs from last lecture.

Suppose Z is irreducible, φ is a rational map, and that $Z \cap$ the locus where φ is defined is nonempty.

Definition 12.3. The **total transform** of Z is $\pi_2(\pi_1^{-1}(Z))$.

The **proper transform** is the closure in Y of $\varphi(Z \cap U)$ where φ is regular on U .

We now consider some basic examples.

Example 12.4. 1. (Projections) Suppose we have a point $p \in \mathbb{P}^n$ and a hyperplane $H \cong \mathbb{P}^{n-1}$, where $p \notin H$. Recall our definition of the projection $\pi_p : \mathbb{P}^n \setminus \{p\} \rightarrow H \cong \mathbb{P}^{n-1}$.

2. (Generalized projection) Consider linear subspace $\Lambda \cong \mathbb{P}^k \subset \mathbb{P}^n$ and a complementary subspace $\Phi \cong \mathbb{P}^{n-k-1} \subset \mathbb{P}^n$. By complementary, we mean that $\Phi \cap \Lambda = \emptyset$ or equivalently Φ, Λ span $\mathbb{P}^n$. Now, we can define $\pi_\Lambda : \mathbb{P}^n \dashrightarrow \Phi$ where we send $p \mapsto \overline{\Lambda, p} \cap \Phi$. In words, if we take any $p \notin \Lambda$, then the span $\overline{\Lambda, p}$ must intersect Φ at exactly one point.

3. (Very specific projection) Take the projection map $\pi_p : \mathbb{P}^2 \dashrightarrow L$ where $q \neq p$ is sent to $\overline{pq} \cap L$.

Question 12.5. What does the graph of our very specific projection π_p look like?

Answer: The graph Γ_π is a spiral staircase. Each line through the origin is mapped to a single point in the codomain (which is isomorphic to $\mathbb{P}^1$).

This is what is known as a “blow up” at the point p . Let's try (against maybe some of our better judgment) to describe blow-ups in greater generality.

Definition 12.6. Take $X \in \mathbb{P}^n$ and a point $p \in X$, choose a hyperplane $H \subset \mathbb{P}^n$ with $H \not\ni p$. We have the projection map $\pi_p : X \dashrightarrow H \cong \mathbb{P}^{n-1}$. We define the **blow-up** of X at p to be the graph of π_p .

Remark 12.7. We can think of the blow-up as replacing the point p by the possible limits of lines joining p to other points on X .

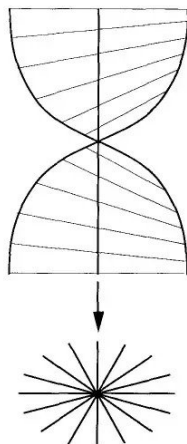


Figure 1: The graph of $\pi_p : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$

Definition 12.8. Take an affine variety $X \subset \mathbb{A}^n$ and a closed subvariety $Z \subset X$ such that $Z = V(f_1, \dots, f_k)$. We can define a rational map $\pi : X \dashrightarrow \mathbb{P}^{k-1}$ which maps $p \mapsto [f_1(p), \dots, f_k(p)]$. This is a rational map that is regular on the complement of Z . Define the **blow-up** $Bl_Z(X) := \Gamma_\pi$ the graph of this map.

One last thing: there's a projective version of the blow-up, which is similar to the above except for a technical difference. Let $X \subset \mathbb{P}^n$ be a variety, with $Z \subset X$ closed. Choose homogeneous polynomials $F_1, \dots, F_k$ of the **same degree** d such that $\text{sat}(F_1, \dots, F_k) = I_Z$. Then we get a rational map

$$\pi : X \dashrightarrow \mathbb{P}^{k-1}$$

where $p \mapsto [F_1(p), \dots, F_k(p)]$. The graph of this map is the blow-up.

12.1 Back to birational isomorphism

Recall the definition of birational isomorphism from last lecture.

Definition 12.9. We say a variety X is **rational** if $K(X) \cong K(Z_1, \dots, Z_n)$, or equivalent it is birationally isomorphic to $\mathbb{P}^n$.

Claim 12.10. If $Q \subset \mathbb{P}^3$

13 Midterm!

14 Lecture 13: 3/12/26

Today, we will finish our discussion of rational maps. When we get back, we're going to start Part 2 of Harris: A First Course, which begins with dimension.

14.1 Birational Review

For now, we say that X is an irreducible variety, and recall the definition of rational maps:

Definition 14.1. A **rational map** $f : X \dashrightarrow Y$ is an equivalence class of pairs (U, f_u) where U is open and $f_u : U \rightarrow Y$ is regular under the equivalence relation $(U, f_u) \sim (V, f_v)$ if $f_u|_{U \cap V} = f_v|_{U \cap V}$. Note that it is important that the varieties are irreducible so that the open sets are dense.

Definition 14.2. Let $\varphi : X \dashrightarrow Y$ be a rational map. Take open $U \subset X$ such that φ is regular on U , then we can form

$$\Gamma_{\varphi, U} \{ (u, \varphi(u)) \} \subset U \times Y \subset X \times Y.$$

Define the **graph of a rational map** $\Gamma_\varphi \subset X \times Y$ to be the closure of $\Gamma_{\varphi, U}$ for any open $U \subset X$.

That is, a rational map $X \dashrightarrow Y$ is a closed subvariety $\Gamma \subset X \times Y$ for which there exists some $U \subset X$ such that the projection onto the first factor $\pi_1^{-1}(U) \cap \Gamma \xrightarrow{\sim} U$ is an isomorphism.

Definition 14.3. We say X, Y are **birational** or **birationally isomorphic** if there exist rational maps $f : X \dashrightarrow Y$ and $g : Y \dashrightarrow X$ such that $f \circ g = \text{Id}_Y$ and $g \circ f = \text{Id}_X$. Alternatively, this means $K(X) \cong K(Y)$. This defines an equivalence relation on varieties.

Equivalently, we can also say that X, Y are birational $\iff$ there exist open $U \subset X, V \subset Y$ such that $U \cong V$.

Most geometric categories do not have such “quasi maps” like algebraic geometry. This birationality gives a coarser equivalence relation than classical isomorphism, which opens the problem of classifying all varieties up to birational isomorphism. This gives the idea of the minimal model program, that attempts to answer the following question: within a specific birational isomorphism class, is there a simplest such variety?

Remark 14.4. We can always compose birational maps $X \xrightarrow{f} Y \xrightarrow{g} Z$. In particular, we see that birational maps $X \dashrightarrow X$ form a group $\text{Bir}(X)$. We ask ourselves: what is $\text{Bir}(\mathbb{P}^1)$? What is $\text{Bir}(\mathbb{P}^2)$? Of course, this includes PGL_{n+1} for $\mathbb{P}^n$, but we want to see if there are others and classify them as a group.

14.2 Reintroducing Blow-ups

We now look at some examples:

Example 14.5. We have the rational map

$$\begin{aligned} \pi : \mathbb{A}^2 &\dashrightarrow \mathbb{P}^1 \\ (x, y) &\longmapsto [x, y] \end{aligned}$$

We want to think of rational maps in terms of their graphs. The graph of this example looks something like an infinite spiral staircase, for which each line through the origin is mapped to a single point in the codomain.

Now take $C \subset \mathbb{A}^2$ defined by $C = V(y^2 - x^2(x+1))$, for which we can take the map π and restrict π to $C \subset \mathbb{A}^2$, which gives a rational map from the curve C to $\mathbb{P}^1$.

These are both examples of blow-ups.

Definition 14.6. Take an affine variety $X \subset \mathbb{A}^n$ and a closed subvariety $Z \subset X$ such that $Z = V(f_0, \dots, f_m)$. We can define a rational map $\pi : X \dashrightarrow \mathbb{P}^m$ which maps $p \mapsto [f_0(p), \dots, f_m(p)]$. This is a rational map that is regular on the complement of Z . We define the **blow-up** $\text{Bl}_Z(X) := \Gamma_\pi$ to be the graph of this map.

In the last couple of decades, it has been proven that in fact every birational map is a composition of blow-ups and blow-downs! This result is known as the Weak Factorization Theorem, proven by Abramovich, Karu, Matsuki, and Włodarczyk in 2002. Fun fact: Dan Abramovich's PhD thesis, titled "Subvarieties of abelian varieties and of Jacobians of curves", was supervised by Joe Harris.

We now want to look at a quadric surface in $\mathbb{P}^3$.

Example 14.7. Let $Q = V(XY - ZW) \subset \mathbb{P}^3$. We know this is the Segre image of $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ in $\mathbb{P}^3$. Now we take the point $p : [0, 0, 0, 1] \in Q$ and consider the projection map

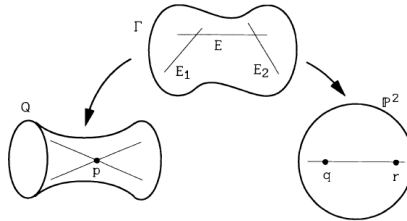
$$\pi_p : Q \dashrightarrow \mathbb{P}^2$$

Now the graph Γ of π_p is exactly $\text{Bl}_{\{p\}}(Q)$. This means we get a regular map $\text{Bl}_{\{p\}}(Q) \rightarrow \mathbb{P}^2$. We ask ourselves: is this map an isomorphism?

No! We see from the following diagram

$$\begin{array}{ccc} & \Gamma & \\ \alpha \swarrow & & \searrow \beta \\ Q & & \mathbb{P}^2 \end{array}$$

that proper transforms of lines $L, M \subset Q$ through the point p are now disjoint, so they are collapsed (they can be blown-down) to points. That is, α is the blow-up of Q at p , and β is the blow-up of $\mathbb{P}^2$ at two points $\pi(L), \pi(M)$, so the picture looks like this:



Definition 14.8. For a variety X and subvariety Z where we blow up the subvariety $Z \subset X$ and look at the map $\pi : \text{Bl}_Z(X) \rightarrow X$, then $E = \pi^{-1}(Z)$ is an **exceptional divisor**.

This explanation is, in Prof. Harris' words, "more for the sake of general culture" as we won't be getting deeply involved in these constructions.

14.3 Back to the Join!

Now we want to look again at the join. We take $X, Y \subset \mathbb{P}^n$ with $X \cap Y = \emptyset$, and we now write the join as

$$\mathcal{J}(X, Y) = \bigcup_{p \in X, q \in Y} \overline{pq}$$

We can now take the same construction as seen before, which gives the diagram

$$\begin{array}{ccc} & \Sigma = \{(L, p) \mid p \in L\} \subset \mathbb{G}(1, n) \times \mathbb{P}^n & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ \mathbb{G}(1, n) & & \mathbb{P}^n \end{array}$$

For which

$$\mathcal{J}(X, Y) = \pi_2(\pi_1(\mathcal{C}_1(X) \cap \mathcal{C}_1(Y)))$$

We note that this equivalently gives a map

$$\begin{aligned} X \times Y &\longrightarrow \mathbb{G}(1, n) \\ (p, q) &\longmapsto \overline{pq} \end{aligned}$$

Now if X, Y do meet, we still get a rational map given by

$$X \times Y \dashrightarrow \mathbb{G}(1, n)$$

Now we can define $J := \text{im}(\varphi) \subset \mathbb{G}(1, n)$, so that

$$\mathcal{J}(X, Y) = \pi_2(\pi_1^{-1}(J))$$

for which we know that since the locus $\mathcal{J}(X, Y) \subset \mathbb{G}(1, n)$ of lines meeting both X and Y is a subvariety of the Grassmannian, then the union $J(X, Y)$ of these lines for which is a subvariety of $\mathbb{P}^n$. Note that this still makes sense when $X = Y$, for which this set is called the secant variety of X , denoted $\text{Sec}(X)$.

15 Lecture 14: 3/24/26

Now that we are back from spring break, we will begin the second half of the course, looking at properties of varieties. We will begin with the notion of dimension.

This is a good example of the “algebraicization” of algebraic geometry. In the classical algebraic geometry, there was no good definition of dimension. It was somewhat a “you know it when you see it”¹ approach. For example, Semple and Roth in 1950 said that a variety has dimension n if it consists of a ∞^n of points.

Remark 15.1. We will define dimension of X for irreducible varieties first. Then, for all varieties $X = \bigcup X_\alpha$ for irreducible X_α , we say $\dim(X) = \max(\dim X_\alpha)$ and the **local dimension** at $p \in X$ is

$$\dim_p X := \max_{\alpha: p \in X_\alpha} \dim X_\alpha$$

¹https://en.wikipedia.org/wiki/I_know_it_when_I_see_it

We have a starting point that we should have $\dim \mathbb{P}^n = n$. We also want to have that a surjective map $X \rightarrow \mathbb{P}^k$ with finite fibers then $\dim X = k$. These two points together do not give a definition, but provide great guidance of where to start. The following facts will help make this into a definition.

Proposition 15.2. *For any $X \subset \mathbb{P}^n$, we claim that there exists a projection $\pi_\Lambda : X \rightarrow \mathbb{P}^k$ which is finite and surjective.*

Proof. Start by picking a point not in X and projecting onto a $n - 1$ dimensional plane. Then, the projection is always finite. If it is not surjective, repeat this process of projecting until the map is surjective. $\square$

Definition 15.3. The dimension of X is the smallest k such that every $(n - k)$ -plane in $\mathbb{P}^n$ meets X . This is also the smallest k such that there exists a $n - k - 1$ disjoint from X .

Exercise 15.4. Check that the definition agrees with the notion from the previous proposition of having a surjective.

Remark 15.5. This is not a bad definition, but this definition is not a priori invariant under isomorphism. So, we need another characterization of dimension.

Remark 15.6. For X irreducible, if $X \rightarrow \mathbb{P}^k$ is a finite surjective map, we have $f^* : K(\mathbb{P}^k) \hookrightarrow K(X)$. Then, the dimension of X is the transcendence degree of $K(X)/K = k$. This is nicely invariant under isomorphisms, which is good.

Theorem 15.7. *This characterization in terms of transcendence degree gives immediately that dimension is invariant under birational maps. So, any irreducible varieties that contain isomorphic open sets have the same dimension.*

Let's do some examples!

Example 15.8. • We have that $\dim \mathbb{P}^n = n$.

- Now, let's consider $\dim G(k, n)$. We have seen in the past that the Grassmanian $G(k, n)$ has an open subset isomorphic to $\mathbb{A}^{k(n-k)}$. So, $\dim G(k, n) = k(n - k)$.

Now, let's get some properties of dimension.

Proposition 15.9. *If we have varieties X, Y , then $\dim(X \times Y) = \dim X + \dim Y$.*

Proof. We start by noting that $\mathbb{P}^m \times \mathbb{P}^n$ contains $\mathbb{A}^m \times \mathbb{A}^n = \mathbb{A}^{m+n}$ as an open subset. So, $\dim(\mathbb{P}^m \times \mathbb{P}^n) = m + n$.

Then, for any X, Y there exist finite surjective maps $f : X \rightarrow \mathbb{P}^m, g : Y \rightarrow \mathbb{P}^n$. Then, $f \times g$ is a surjective finite map from $X \times Y \rightarrow \mathbb{P}^m \times \mathbb{P}^n$. $\square$

Definition 15.10. Recall that for $X \subset \mathbb{P}^n$ and $p \notin X$, we have the cone:

$$\overline{pX} = \bigcup_{q \in X} \overline{pq}.$$

Proposition 15.11. *If X has dimension k , then $\overline{pX}$ has dimension $k + 1$.*

In order to state our main theorem, we need the following definition:

Definition 15.12. A function $f : X \rightarrow \mathbb{Z}$ is **upper semi-continuous** if $f^{-1}([k, \infty))$ is closed.

Theorem 15.13. Let X be a quasi-projective variety and $\pi : X \rightarrow \mathbb{P}^n$ any map. Let Y be the closure of the image of X . So, we could say $\pi : X \rightarrow Y \subset \mathbb{P}^n$. Then, for all $p \in X$, let $X_p := \pi^{-1}(\pi(p))$. So, X_p is a variety for all p . Set $\mu_p := \dim_p(X_p)$. So, μ_p is an integer valued function on X .

Then, μ is upper semi-continuous and if X is irreducible, then $\mu = \min_{p \in X} \mu(p)$ then $\dim(X) = \dim(Y) + u$.

Example 15.14. Let X be the blow up at p of $\mathbb{P}^2$. So,

$$X = \{(q, \ell) \in \mathbb{P}^2 \times \mathbb{P}^{2*} : p, q \in \ell\}.$$

We have a projection onto the first factor where $X_q = \begin{cases} \{q\} & q \neq p \\ \mathbb{P}^1 & q = p. \end{cases}$ So, X has dimension equal to $\dim(\mathbb{P}^2) = 2$.

Let's look at a topological example where this theorem is clearly false.

Example 15.15. Consider the sphere $x^2 + y^2 + z^2 = 1$ for real x, y, z . Then, consider the map that projects on the x -axis. Then, our fibers are circles for $x \in (-1, 1)$ and points for $x \in \{-1, 1\}$. So, the function μ is not upper-semicontinuous: the dimension drops down at the points $x \in \{-1, 1\}$. This problem stems from the fact that we are working over the reals which are not algebraically closed. If we looked at the same example in $\mathbb{C}$, these problems would go away as the fiber over 1 (or -1) is the union of lines, and thus has dimension 1.

Question 15.16. Does every surface $S \subset \mathbb{P}^3$ of degree d contain a line?

Answer: For $d = 1, 2$, we know that the answer is yes! For $d > 2$, we are unsure. So, let's introduce an incidence correspondence. Set $N := \binom{d+3}{3} - 1$. So, we can think of $\mathbb{P}^N$ as the surfaces of degree d in $\mathbb{P}^3$. Consider $\mathbb{G}(1, 3)$, the family of lines in $\mathbb{P}^3$. Set

$$\Phi := \{(S, L) \in \mathbb{P}^N \times \mathbb{G}(1, 3) : L \subset S\}.$$

We have the projection maps:

$$\begin{array}{ccc} & \Phi & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ \mathbb{P}^N & & \mathbb{G}(1, 3) \end{array}$$

If $\dim \Phi < N$, then $\pi_1(\Phi)$ is a proper subvariety of $\mathbb{P}^N$ and thus not every degree d hypersurface contains a line. So, let's calculate $\dim \Phi$. We only have one theorem about dimension, so let's use it.² We have a map

$$\text{polynomials of degree } d \text{ on } \mathbb{P}^3 \rightarrow \{\text{polynomials of degree } d \text{ on } L\}$$

²“When all you have is a hammer, everything starts looking like a nail” -Professor Harris

This map is surjective and calculating degrees, we get that $\pi_2^{-1}(L) \cong \mathbb{P}^{N-d-1}$ for all L . So, applying our theorem, we have $\dim \Phi = N - d + 3$. When $d = 1, 2$ we have that $\dim \Phi > N$, which agrees with our previous notions about surfaces. When $d = 3$, we are unsure. But for $d \geq 4$, we have that $\dim \Phi < N$, and thus the general surface of degree 4 or greater does not contain a line.

16 Lecture 15: 3/26/26

For an irreducible variety X , recall our definitions of dimension and our main theorem (15.13) about dimensions.

We can extend Theorem 15.13 slightly.

Theorem 16.1. *Let X be an irreducible projective variety and $\pi : X \rightarrow Y$ surjective. Then, if we define $\mu(p) := \dim_p(\pi^{-1}(\pi(p)))$ and $\lambda(q) := \dim \pi^{-1}(q)$, both μ and λ are upper semi-continuous.*

Corollary 16.2. *If we let $\mu = \min_{p \in X} \mu(p)$ and $\lambda = \min_{q \in Y} \lambda(q)$. Then, $\dim X = \dim Y + \mu = \dim Y + \lambda$.*

This will all be clearer after a few examples, but the thing to keep in mind is that the set of points where μ or λ is minimized is a dense open set.

16.1 Complete Intersections

Definition 16.3. Let X be a projective variety of dimension k . We say X is a **complete intersection** if $I(X) = (F_1, \dots, F_{n-k})$ for some polynomials $F_1, \dots, F_k$.

Definition 16.4. A projective variety X is a **set-theoretic complete intersection** if $X = V(G_1, \dots, G_{n-k})$ for polynomials $G_1, \dots, G_{n-k}$.

Example 16.5. Consider three non-collinear points $p_1, p_2, p_3 \in \mathbb{P}^2$. For the ease of the example, take p_i to be the coordinate points. We claim that $X = \{p_1, p_2, p_3\}$ is not a complete intersection. We see that $I(X)$ contains the points XY, XZ, YZ and no linear functions vanishes on all three points. So, we need at least three generators to generate $I(X)$, and thus X is not a complete intersection.

On the other hand, X is a set-theoretic complete intersection. We can see this by taking a conic through the three points as well as the zero locus XYZ , which is the union of the three lines connecting the points.

16.2 Examples for dimension

Example 16.6. Let

$$M = \mathbb{P}^{mn-1} = \{m \times n \text{ nonzero matrices mod scalars}\}.$$

Alternatively, $\mathbb{P}^{mn-1} = \mathbb{P}\text{Hom}(K^m, K^n)$. We have the subspaces $M_r \subset M$ of matrices with rank at most r . Alternatively, M_r is the zero locus of the $(r+1) \times (r+1)$ minors of the matrix and is thus a projective variety. As will be a very common question going forward, we want to calculate the dimension of M_r . This will be a quite instructive example.

We want to form an incidence correspondence. Define $\Phi := \{(A, \Lambda) \in M \times G(m-r, m) : \Lambda \subset \ker A\}$. Naturally, we have the following diagram:

$$\begin{array}{ccc} & \Phi & \\ \swarrow \pi^1 & & \searrow \pi_2 \\ M_r & & G(m-r, m) \end{array}$$

To start, we know $G(m-r, m)$ has dimension $r(m-r)$. We also can compute the fibers of π_2 . These are the matrices that vanish on some fixed $m-r$ dimensional subspace λ . This fiber is exactly

$$\mathbb{P}(\text{Hom}(K^m/\Lambda, K^n) \cong \mathbb{P}(\text{Hom}(K^r, K^n) \cong \mathbb{P}^{rn-1}$$

and thus the fiber always has dimension $(m-r)n-1$. We can conclude that Φ has dimension $r(m-r) + rn - 1$. Now, all that remains is to look at the map π_1 . It suffices to consider the fiber over elements of M_r of exactly rank r , as these form an open subset of M_r . Over such points, the fiber is a single point, as there is a unique $m-r$ plane in the kernel, as the kernel is of dimension $m-r$. So, the generic fibers of π_1 have dimension 0 so M_r has dimension $r(m-r) + rn - 1$.

As a note, in order to say that Φ is irreducible in the above example, we need the following theorem.

Theorem 16.7. *Let X be a variety and Y an irreducible variety of dimension k such that there exists a map $\pi : X \rightarrow Y$ such that the fibers $\pi^{-1}(y)$ are irreducible of dimension d for all y . Then, X is an irreducible variety of dimension $k + d$.*

In our remaining time, let's do another example.

Example 16.8 (Universal k -plane). Recall the following construction

$$\begin{array}{ccc} & \Sigma := \{(\Lambda, p) \in \mathbb{G}(k, n) \times \mathbb{P}^n : p \in \Lambda\} & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ \mathbb{G}(k, n) & & \mathbb{P}^n \end{array}$$

Let's calculate some dimensions of fibers. Observe that all of the fibers of π_1 are isomorphic to $\mathbb{P}^k$, as it is exactly the set of points in the fixed Λ . So, from the previous theorem, we have that Σ is irreducible of dimension $(k+1)(n-k) + k$. We could have done this on the other side by noting that the fibers of π_2 are isomorphic to $\mathbb{G}(k-1, n-1)$.

Example 16.9. Let $X \subset \mathbb{P}^n$ be a variety of dimension m and take some $k < n-m$. Define

$$\mathcal{C}_k := \{\Lambda \in \mathbb{G}(k, n) : \Lambda \cap X \neq \emptyset\}.$$

What is the dimension of $\mathcal{C}_k$? Using the incidence correspondence from the previous example, we have that $\mathcal{C}_k = \pi_1(\pi_2^{-1}(X))$. As stated above, the fibers of π_2 over a point p is isomorphic to $\mathbb{G}(k-1, n-1)$. The general fiber of π_1 from $\pi_2^{-1}(X)$ has only a single point. So, $\dim \mathcal{C}_k = m + k(n-k)$.

17 Lecture 16:3/31/26

Looking forward, the remaining topics of the course are:

- Dimension
- Hilbert functions/Hilbert polynomials
- Smoothness, singularity, tangent spaces, tangent cones
- Degree
- Parameter spaces/moduli spaces

So, today we will wrap up the notion of dimension with one last example.

17.1 A last example on dimension

Example 17.1. Recall that for two disjoint varieties $X, Y \subset \mathbb{P}^n$ of dimensions k, ℓ , we have the join

$$J(X, Y) := \bigcup_{p \in X, q \in Y} \overline{pq}.$$

What is the dimension of $J(X, Y)$?

We have a map $s : X \times Y \rightarrow \mathbb{G}(1, n)$ mapping $(p, q) \mapsto \overline{pq}$. Denote the image of this map by $\mathcal{J}(X, Y) \subset \mathbb{G}(1, n)$. Now, as per usual, we have the following incidence correspondence:

$$\begin{array}{ccc} & \Sigma := \{(\ell, p) \in \mathbb{G}(1, n) \times \mathbb{P}^n : p \in \ell\} & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ \mathbb{G}(1, n) & & \mathbb{P}^n \end{array}$$

With this, we can write $J(X, Y) = \pi_2(\pi_1^{-1}(\mathcal{J}(X, Y)))$.

Claim: If X and Y are disjoint, then $s^{-1}(L)$ is finite. To see this, note that if the preimage is positive dimensional, this means that there is a line contained in X or Y that intersects the other, but by assumption the varieties are disjoint.

This means that $\dim(\mathcal{J}(X, Y)) = k + \ell$ which in turn means that $\dim \pi_1^{-1}(\mathcal{J}(X, Y)) = k + \ell + 1$. Since X, Y are disjoint, π_2 is finite (prop 11.37 in the book). So, we have that $\dim J(X, Y) = k + \ell + 1$.

We can generalize this construction to the case $X \cap Y \neq \emptyset$, as we have mentioned earlier in course, by taking pairs of distinct points, which we can then take the closure of.

In the special case when $X = Y$, we have the following example:

Example 17.2. We have a rational map $s : X \times X \dashrightarrow \mathbb{G}(1, n)$ sending $(p, q) \mapsto \overline{pq}$ defined on $p \neq q$. Let the image be denoted by $\mathcal{S}(X) \subset \mathbb{G}(1, n)$. Now, we define the **secant variety** of X to be $\pi_2(\pi_1^{-1}(\mathcal{S}(X)))$ (where these projection maps are from the incidence correspondence in the previous example).

If we just apply the result of the previous example, we expect $S(X)$ to have dimension $2k + 1$ where $k = \dim X$.

Question 17.3. When is $\dim S(X)$ equal to $2k + 1$?

Well we have some partial answers/examples.

Exercise 17.4. If X is an irreducible curve not contained in a plane, then $\dim S(X) = 3$.

This is a fun case to think about. Now, let's hop up a dimension.

Theorem 17.5. *If $X \subset \mathbb{P}^n$ is an irreducible non-degenerate surface and $n \geq 5$ (so we have “room” for the secant variety), then $\dim S(X) = 5$ unless X is the Veronese surface: $v_2(\mathbb{P}^2) \subset \mathbb{P}^5$. In fact $\dim S(v_2(\mathbb{P}^2)) = 4$.*

We will do the dimension calculation for the Veronese surface and omit the proof of the statement in full, as it is quite difficult.

Example 17.6. Let $X = v_2(\mathbb{P}^2)$. We have $\mathcal{S}(X) \subset \mathbb{G}(1, 5)$ is 4-dimensional. So, $\pi_1^{-1}\mathcal{S}(X) \subset \Sigma \subset \mathbb{G}(1, 5) \times \mathbb{P}^5$ has dimension 5. The only remaining part of the calculation is to consider the fibers of the map $\pi_2 : \pi_1^{-1}(\mathcal{S}(X)) \rightarrow \mathbb{P}^5$. We claim that these fibers are always positive dimensional. This means that any point that lies on a secant line actually lives on a 1-parameter family of secant lines of X .

Consider some point $r \in S(X)$. So, there exist $p, q \in X$ such that $r \in \overline{pq}$. For slight abuse of notation, let p, q denote their respective preimages in $\mathbb{P}^2$. Let L be the line connecting p, q in $\mathbb{P}^2$. Under the embedding, L maps to a planar conic. Then, r lies in the same plane as $v_2(L)$, and thus any line through r in this plane intersects the plane conic, and thus intersects X . So, we have a 1-parameter family of lines through r , and thus $\pi_2 : \pi_1^{-1}(\mathcal{S}(X)) \rightarrow \mathbb{P}^5$ has one dimensional fibers, and thus $\dim S(X) = 4$.

Question 17.7. *What is $S(X)$?*

17.2 A little bit on smoothness

We take $X \subset \mathbb{P}^n$ and $p \in \mathbb{P}^n$, for which we can take the projection $\pi_p : X \rightarrow \mathbb{P}^{n-1}$. We want to ask naively when this projection map is an isomorphism with its image.

Proposition 17.8. *If X is smooth, the projection map π_p is an isomorphism of X with its image in $\mathbb{P}^{n-1}$ if and only if $p \notin S(X)$.*

Remark 17.9. The first direction of this is clear. If $p \in S(X)$, then p lies on some line connecting points in X , and those points will be projected onto the same point.

The conclusion of this line of reasoning is that any smooth, k -dimensional projective variety X is embeddable in $\mathbb{P}^{2k+1}$.

17.3 Hilbert functions and Hilbert polynomials

Before we get started, we need some setup:

Let $X \subset \mathbb{P}^n$ be an arbitrary projective variety. Let $S := K[Z_0, \dots, Z_n]$ be the coordinate ring of $\mathbb{P}^n$. This is a graded ring, i.e. $S = \bigoplus S_d$ where S_d is the homogeneous polynomials of degree d . Let $I(X)$ denote the homogeneous ideal of X and $I(X)_d = I(X) \cap S_d$. Finally, we have $S(X) = S/I(X) = \bigoplus S_d(X)$.

While S and $S(X)$ are huge rings, the degree d components are finite dimensional vector spaces.

Definition 17.10. The **Hilbert Function** $h_X : \mathbb{N} \rightarrow \mathbb{N}$ of X is the function on natural numbers³ defined by $h_X(d) = \dim S(X)_d$.

Remark 17.11. Note that this function depends on both the variety X and the choice of embedding.

Theorem 17.12. *There exists a polynomial $p_X(d)$ such that for all d sufficiently large, $h_X(d) = p_X(d)$. We call p_X the **Hilbert Polynomial** of X .*

Example 17.13. Let $X = \{p, q, r\} \subset \mathbb{P}^2$. Let's calculate h_X . To start, note that $h_X(0) = 1$. Let's compute $h_X(2)$. Let $S = K[X, Y, Z]$ and note that $I(X)_2$ is the polynomials of degree 2 vanishing on X . We have that $S(X)_2 = S_2/I(X)_2$. Now, noting that vanishing on the points impose independent conditions on the space of polynomials. Using this logic, we see that $h_X(d) = 3$ for all $d \geq 2$.

It remains to calculate $h_X(1)$. In fact, we have not given enough information to determine this. If the three points are collinear, $h_X(1) = 2$. Otherwise, $h_X(1) = 3$.

Exercise 17.14. If $X = \{p_1, \dots, p_m\} \subset \mathbb{P}^2$, then $p_X(d) = m$. More precisely, for $d \geq m - 1$, $h_X(d) = p_X(d) = m$.

³We will take 0 to be a natural number

18 Lecture 18: 4/2/26

Recall that for $X \subset \mathbb{P}^n$ a projective variety, we have $S = S(\mathbb{P}^n) = K[Z_0, \dots, Z_n]$. Inside of this ring, we have the graded ring $I(X)$. Recall that we have the graded homogeneous coordinate ring $S(X) := S/I(X)$. Recall the definition of the Hilbert function and polynomial.

Remark 18.1. The Hilbert function is also the codimension in space S_n of homogeneous polynomials on $\mathbb{P}^n$ of the subspace $I(X)_m$. Equivalently, this is the number of conditions imposed by X on homogeneous polynomials of degree m .

Example 18.2. Recall that for a finite set of points $X = \{p_1, \dots, p_d\} \subset \mathbb{P}^n$ and $m > d - 1$, we have $h_X(m) = d$.

Question 18.3. For a finite set of d points, what are the possible functions $h_X(m)$ for $m \in [1, d - 2]$?

This question is still open, although much progress has been made.

Let's now look at the next example.

Example 18.4. Let $X \subset \mathbb{P}^2$ be a plane curve of degree d : i.e. $I(X) = (F)$ for some irreducible homogeneous polynomial of degree d in S .

Then,

$$\dim I(X)_m = \begin{cases} 0 & m < d \\ \binom{m-d+2}{2} & m \geq d \end{cases}$$

Correspondingly, the Hilbert function is

$$h_X(m) = \begin{cases} \binom{m+2}{2} & m < d \\ \binom{m+2}{2} - \binom{m-d+2}{2} & m \geq d \end{cases}.$$

Observe that $h_X(m)$ is a polynomial for sufficiently large m .

We will now prove the following theorem stated last class.

Theorem 18.5. *There exists a polynomial $p_X(m)$ such that $h_X(m) = p_X(m)$ for all $m \gg 0$. We call p_X the Hilbert polynomial.*

Before we get to this proof, let's do a couple more examples.

Example 18.6. Let $X = L \cup M \subset \mathbb{P}^3$ where L, M are skew lines. We know that $h_X(0) = 1$.

Claim: for $m \geq 1$, we have an exact sequence

$$0 \rightarrow I(X)_m \rightarrow S_m \rightarrow S(X)_m \rightarrow 0.$$

Calculating dimensions on this exact sequence, we can get the dimensions of all the necessary parts. So,

$$h_X(m) = \begin{cases} 1 & m = 0 \\ 2m + 2 & m > 0 \end{cases}$$

If the lines met, the $2m + 2$ becomes $2m + 1$, as we can pick $2m + 1$ points on the two lines (including the point of intersection) such that vanishing on the union of lines is equivalent to vanishing on those $2m + 1$ points.

One last example (for now)!

Example 18.7. Let X be the twisted cubic in $\mathbb{P}^3$. Then, we again we have an exact sequence

$$0 \longrightarrow I(X)_m \longrightarrow S_m \longrightarrow S(X)_m \longrightarrow 0.$$

Now, the twisted cubic is the image of the map $[X, Y] \rightarrow [X^3, X^2Y, XY^2, Y^3]$. So, we have $S(X)_m \cong S(\mathbb{P}^1)_{3m}$ and these are both of dimension $3m + 1$. So, $h_X(m) = p_X(m) = 3m + 1$.

We have now done enough examples. Let's prove something!

Proof of Theorem 18.5. We will prove this by induction. Take some arbitrary variety X . There exists some hyperplane $H = V(L) \subset \mathbb{P}^n$ such that for $Y := X \cap H$, we have $I(Y) = \text{sat}(I(X), L)$. This means that $I(X), L$ cut out Y scheme theoretically. Equivalently, this means that $(I(X), L)_m = I(Y)_m$ for all sufficiently large m . This gives us the following exact sequence:

$$0 \longrightarrow S(X)_{m-1} \longrightarrow S(X)_m \longrightarrow S(Y)_m \longrightarrow 0.$$

So, for sufficiently large m , we have $h_Y(m) = h_X(m) - h_X(m-1)$.

Now, for any function $f : \mathbb{N} \rightarrow \mathbb{N}$, define $f^{(k)}$ to be the k th difference function of f .

Going back to the equation we have above, if we take the intersection $k = \dim X$ times, we see that for sufficiently large m , we see that $h_X^{(k)}$ is constant, meaning that h_X is polynomial. $\square$

Remark 18.8. We can take a quick algebraic diversion. Let S be a graded ring. We say a module M/s is graded is $M = \oplus M_m$ where $S_\ell M_m \subset M_{\ell+m}$.

We have the following piece of language: For a graded module M , define a **twist** $M(\ell)$ defined by $M(\ell)_m = M_{\ell+m}$. This is just a relabeling of the graded pieces.

Take a graded finitely generated S -module M with generators f_1, f_c such that $\deg f_\alpha = d_\alpha$. Then, there exists a surjection of graded S -modules:

$$\bigoplus_{\alpha=1}^c S(-d_\alpha) \longrightarrow M \longrightarrow 0$$

given by multiplication by $(f_1, \dots, f_c)$. So, to understand M , it suffices to understand the kernel of the above map. This kernel is again a finitely generated S -module. We can rinse and repeat this process in order to get a long exact sequence

$$\cdots \longrightarrow \bigoplus S(-d_\beta) \longrightarrow \bigoplus_{i=1}^c S(-d_\alpha) \longrightarrow M \longrightarrow 0$$

The remarkable fact is that this process will eventually terminate with a zero on the left side. Due to Hilbert's syzygy theorem, we have that if S is a polynomial ring over n variables, this sequence terminates after n iterations.

Now, we can state the punchline. We can use the fact that alternating sum of dimensions in an exact sequence is always zero. Using this fact on the graded pieces of M , we get that for sufficiently large M , these dimensions are all polynomials, as the dimension of each graded piece of the direct sums of $S(-d_\alpha)$ are all polynomials as well.

It may be a fun exercise to work out what this looks like for some small examples, such as a collection of points or the twisted cubic.

19 Lecture 19: 4/7/26

We have a couple of announcements. Homework 8 will be due on the 17th of April, and Homework 9 on the 27th of April. The final exam is on the 12th of May.

19.1 Some examples of resolutions

Recall from last lecture our definition of graded modules, and our exposition of the Hilbert Syzygy Theorem.

Let $S = K[Z_0, \dots, Z_n]$. We have

$$S_d = \binom{n+d}{n}$$

where we specifically define the binomial coefficient to be

$$\binom{n}{k} := \begin{cases} \frac{n(n-1)\dots(n-k+1)}{k!} & \text{when } n \geq k \\ 0 & \text{when } k > n \end{cases}$$

From the Syzygy theorem, it follows that $h_M(d) := \dim M_d$ is a polynomial in d for $d \gg 0$.

Example 19.1. Let Γ be three non-collinear points in $\mathbb{P}^2$, which we can transform to $\{[1, 0, 0], [0, 1, 0], [0, 0, 1]\}$. Then I_Γ is generated by (XY, XZ, YZ) . We have a surjection

$$S(-2)^{\oplus 3} \rightarrow I_\Gamma \rightarrow 0$$

Equivalently, we have an exact sequence

$$S(-2)^{\oplus 3} \rightarrow S \rightarrow S_\Gamma \rightarrow 0$$

where the leftmost map is given by $(XY, XZ, YZ) = (F_1, F_2, F_3)$. The kernel of this map is given by the linear relations $ZF_1 = YF_2$ and $YF_2 = XF_3$. So, we extend our sequence to the left, getting the exact sequence

$$0 \longrightarrow S(-3)^{\oplus 2} \xrightarrow{\phi} S(-2)^{\oplus 3} \xrightarrow{(F_1, F_2, F_3)} S \longrightarrow S_\Gamma \longrightarrow 0$$

$$\text{where } \phi = \begin{pmatrix} Z & 0 \\ -Y & Y \\ 0 & -X \end{pmatrix}.$$

Example 19.2. Consider now the twisted cubic, defined by equations F_1, F_2, F_3 . We get an exact sequence

$$S(-2)^{\oplus 3} \rightarrow S \rightarrow S_C \rightarrow 0$$

where the leftmost map is (F_1, F_2, F_3) . The kernel of this map has generators $Z_0F_3 - Z_1F_2 + Z_2F_1 = 0 = Z_1F_3 - Z_2F_2 + Z_3F_1$. We can extend the sequence by one more to

the left, as above, at which point the process will terminate (we'll have an injection on the leftmost side).

19.2 A Motivation for Tangent Spaces

We can now ponder a basic question which has nothing to do with Algebraic Geometry.

Question 19.3. *How do we define the tangent space of a manifold?*

This will be background and motivation for the definition of the Zariski tangent space, which will be a direct analog to what we do with smooth manifolds. We would like to consider tangent vectors as equivalence classes of arcs. For any $p \in X$, an arc through p is a differentiable map $\gamma : (-\epsilon, \epsilon) \rightarrow X$, where $\gamma(0) = p$. We impose the equivalence relation that $\gamma \sim \delta$ if $\gamma'(0) = \delta'(0)$. Heuristically, that means that near 0 these arcs look the same.

Alternatively, let $\mathcal{O}_{x,p}$ be the ring of germs of smooth functions on neighbourhoods U of $p \in X$. Concretely, we define

$$\mathcal{O}_{x,p} = \{(U, f) : p \in U \subset X, f \in \mathcal{C}^\infty \text{ on } U\}$$

modded out by the equivalence relation $(U, f) \sim (V, g)$ if f, g agree on $U \cap V$. This is a local ring, with maximal ideal $\mathfrak{m}_{X,p} = \{(U, f) | f(p) = 0\}$. What's the point of this? A tangent arc γ gives rise to a linear form on $\mathfrak{m}_{X,p}$ via the composition $f \mapsto (f \circ \gamma)'(0)$. By the product rule, this factors through $\mathfrak{m}_p/\mathfrak{m}_p^2$. This gives an identification

$$T_{X,p} \cong (\mathfrak{m}_p/\mathfrak{m}_p^2)^*$$

This is the definition of the Zariski tangent space.

20 Lecture 20: 4/9/26

To repeat: Homework 8 will be due on the 17th of April, and Homework 9 on the 27th of April. The final exam is on the 12th of May.

20.1 Zariski tangent space

Today, we will begin by talking about tangent spaces to varieties. Take any variety X , and some point $p \in X$. We define as before:

Definition 20.1. The **local ring** is the ring of germs of regular functions in neighborhoods of p . We can write this as

$$\mathcal{O}_{x,p} = \{(U, f) \mid p \in U \subset X \text{ open, } f \text{ regular function on } U\} / \sim$$

where the equivalence relation is given by

$$(U, f) \sim (V, g) \text{ if } f|_{U \cap V} = g|_{U \cap V}$$

This is a local ring with a maximal ideal $\mathfrak{m}_p := \{(U, f) \mid f(p) = 0\}$.

Concretely, if $X \subset \mathbb{A}^n$, then we have that $A(X)$ is the coordinate ring of X , given by $K(z_1, \dots, z_n)/I(X)$. Then we take the maximal ideal $\mathfrak{m}$ of f , for which $f(p) = 0$. Then

$$\mathcal{O}_{X,p} = A(X)_{\mathfrak{m}_p}$$

That is, in the language of commutative algebra, the local ring $\mathcal{O}_{X,p}$ the localization of the coordinate ring $A(X)$ at the maximal ideal $\mathfrak{m}_p$.

Definition 20.2. We now have a definition of the **Zariski tangent space**:

$$T_p(X) := (\mathfrak{m}_p/\mathfrak{m}_p^2)^*$$

Note also that there are situations where these constructions might lead to disaster. Take, for example, a topological manifold, where we can think about the same local ring of germs of continuous functions. Think about what we would get if we were so foolish as to go ahead and do this. It is a vector space, but it's not finite dimensional – it's not even clear that this is vector space is of countable dimension. That is to say, this construction works in broader settings, but not in every setting.

We now look at the key property of the following regular map:

$$\begin{aligned} \phi : X &\longrightarrow Y \\ p &\longmapsto q \end{aligned}$$

We set the pullback map to be

$$\begin{aligned} \phi^* : \mathcal{O}_{Y,p} &\longrightarrow \mathcal{O}_{X,p} \\ (U, f) &\longmapsto (\phi^{-1}(U), \phi \circ f) \end{aligned}$$

This pullback map ϕ^* takes $\mathfrak{m}_q$ to $\mathfrak{m}_p$, and similarly $\mathfrak{m}_q^2$ to $\mathfrak{m}_p^2$. This gives us the induced map

$$\mathfrak{m}_q/\mathfrak{m}_q^2 \rightarrow \mathfrak{m}_p/\mathfrak{m}_p^2$$

Definition 20.3. The transpose of the above induced map is the **differential**:

$$d\phi : T_p(X) \rightarrow T_q(Y)$$

20.2 Smooth and singular

This leads us to some key definitions:

Theorem 20.4. Assume X is irreducible, and take $p \in X$. Then

$$\dim T_p(X) \geq \dim X$$

Definition 20.5. If $\dim T_p(X) = \dim X$ we say p is a **smooth** point of X .

Definition 20.6. If $\dim T_p(X) > \dim X$ we say p is a **singular** point of X .

Definition 20.7. We say X is **smooth** if X is smooth at every $p \in X$.

We denote

$$X_{\text{sm}} := \text{locus of smooth pts of } X$$

$$X_{\text{sing}} := \text{locus of singular pts of } X$$

We note that X_{sm} is open and nonempty, and that X_{sing} is a proper closed subset of X .

Concretely, if $X \subset \mathbb{A}^n$ with pure dimension l , and we have $I(X) = (f_1, \dots, f_k) \subset K[x_1, \dots, x_n]$, then we can let

$$M = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \cdots & \vdots \\ \frac{\partial f_k}{\partial x_1} & \cdots & \frac{\partial f_k}{\partial x_n} \end{pmatrix}$$

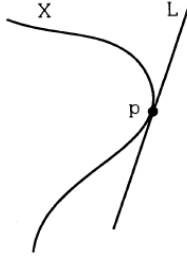
This gives that X_{sm} is the locus where $\text{rank } M = n - l$, and X_{sing} is the locus where $\text{rank } M < n - l$.

Example 20.8. If $X = V(f) \subset \mathbb{A}^n$ hypersurface, then

$$X_{\text{sing}} = V\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) \cap X = V\left(f, \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right)$$

20.3 Affine and projective tangent spaces

Now say $p \in X \subset \mathbb{A}^n$. There arises some confusion, since we like to draw pictures like below and call the line L in question the “tangent space to X at p ”. In reality, the line drawn is



what we will call the **affine tangent space to X at p** . That is, the affine subspace of $\mathbb{A}^n$ through p parallel to $T_p(X)$.

Now suppose instead that $X \subset \mathbb{P}^n$ is a projective variety, with $p \in X$. Assume also that $p \in \mathbb{A}^n \subset \mathbb{P}^n$. This gives the following seemingly complicated construction:

Definition 20.9. The **projective tangent space to X at p** , denoted $\mathbb{T}_p(X) \subset \mathbb{P}^n$, is given by taking the affine tangent space to $X \cap \mathbb{A}^n$ and taking its closure in $\mathbb{P}^n$. We can alternatively say that for $I_X = (F_1, \dots, F_l)$ with $F_i \in K[z_0, \dots, z_n]$, we have

$$\mathbb{T}_p(X) := \left\{ [Z_0, \dots, Z_n] \mid \sum_i Z_i \frac{\partial F_\alpha}{\partial Z_i}(p) = 0 \ \forall \alpha \right\}$$

Remark 20.10. For example, if $X = V(F)$, then:

- If $\frac{\partial F}{\partial Z_\alpha}(p) = 0 \forall \alpha$, then p is a singular point of X and $\mathbb{T}_p(X) = \mathbb{P}^n$.
- If $\frac{\partial F}{\partial Z_\alpha}(p)$ are not all 0, then p is a smooth point of X , and $\mathbb{T}_p(X)$ is the zero locus of linear form $[z_0, \dots, z_n] \mapsto \sum_i Z_i \frac{\partial F}{\partial Z_i}(p)$.

Appreciate that when people talk about the tangent space or the tangent plane, you never really know what they mean.

Example 20.11. We take the simplest case of $X = V(f) \subset \mathbb{P}^2$, with $F = F(X, Y, Z)$ homogeneous of degree d . We also take $p = [X_0, Y_0, Z_0] \in X \subset \mathbb{P}^2$. This gives

$$\mathbb{T}_p(X) = \left\{ [X, Y, Z] \mid X \cdot \frac{\partial F}{\partial X}([X_0, Y_0, Z_0]) + Y \cdot \frac{\partial F}{\partial Y}([X_0, Y_0, Z_0]) + Z \cdot \frac{\partial F}{\partial Z}([X_0, Y_0, Z_0]) = 0 \right\}$$

Notice that $p \in X$ is singular if

$$F(p) = \frac{\partial F}{\partial X}(p) = \frac{\partial F}{\partial Y}(p) = \frac{\partial F}{\partial Z}(p) = 0$$

In characteristic 0, this is redundant. This gives the **Euler relation**. If F is homogeneous of degree d , then

$$dF = X \frac{\partial F}{\partial X} + Y \frac{\partial F}{\partial Y} + Z \frac{\partial F}{\partial Z}$$

20.4 Gauss maps

We now begin talking about Gauss maps, which we will discuss in more detail next week.

Example 20.12. We start with the case $X = V(F) \subset \mathbb{P}^2$. Our previous constructions of the day give us a regular map

$$\begin{aligned} X &\longrightarrow (\mathbb{P}^2)^* \\ p &\longmapsto \mathbb{T}_p(X) \\ p &\longmapsto \left[\frac{\partial F}{\partial X}, \frac{\partial F}{\partial Y}, \frac{\partial F}{\partial Z} \right] \end{aligned}$$

To clarify this example of the Gauss map, we can look at the curve $ZY^2 - X^2(X - Z)$, which has a singular point. When we look at the Gauss map of this curve, we see that it is in fact not regular at the singular point p , since the limiting tangent lines differ.

Keep in mind that this is a very, very special case of the much more general construction of Gauss maps. We will now provide the definition, to be reiterated next week.

Definition 20.13. In general, if $X \subset \mathbb{P}^n$ is a smooth, pure k -dimensional variety, then we are associating a linear subspace of the projective tangent space to every point in X . That is,

$$p \in X \rightsquigarrow \mathbb{T}_p(X) \cong \mathbb{P}^k \subset \mathbb{P}^n$$

This gives a regular map

$$\mathcal{G} = \mathcal{G}_X : X \longrightarrow \mathbb{G}(k, n)$$

which we call the **Gauss map**. Note that if X is singular, then this is a rational map.

We will cover many more constructions like this in the coming weeks. In the meantime, let us talk about one of the major problems in algebraic geometry – birational geometry and rational maps. This is the focus of the **minimal model program**. The crucial question of this setting: is every projective variety birational to a smooth projective variety?

The stakes are quite high, since we can see that there are a lot of nice properties and conditions that smooth varieties have and singular varieties lack. This question was partially answered by Heisuke Hironaka in the 1960s, who proved a wonderful theorem on the resolution of singularities. Hironaka, who passed away this past month, won a Fields medal for his work on this problem in characteristic 0.

21 Lecture 21: 4/14/26

Recall from last week the definition of the Zariski tangent space $T_p X$ of a variety X at a point $p \in X$. Recall the definitions of smooth and singular points.

Remark 21.1 (General Facts). Here are some results about the tangent space.

- $\dim T_p X \geq \dim X$ is the only inequality holding in general. For example, we could take X to be the union of coordinate axes in $\mathbb{A}^n$. In this case, at $p = 0$, $\dim T_p X = n$. By taking product with $\mathbb{A}^k$, we see that we can construct tangent spaces of any dimension greater than $\dim_p X$.
- $\dim T_p X$ is an isomorphism invariant. This gives us the following unintuitive example: $X = V(xy, yz, xz) \subset \mathbb{A}^3$ and $Y = V(xy(x - y)) \subset \mathbb{A}^2$ are not isomorphic as varieties, despite both being the union of three concurrent lines.
- $X_{sm} \subset X$ is an open and dense subset and X_{sing} is a proper closed subset.
- If X, Y are projective varieties and $f : X \rightarrow Y$ is an isomorphism if and only if f is bijective and $df : T_p X \rightarrow T_{f(p)} Y$ is injective for all $p \in X$. There is a substantially more general statement about quasiprojective varieties in the book.

Example 21.2 (Cuspidal Cubic). Consider $C := V(Y^2 Z - X^3) \subset \mathbb{P}^2$. (in affine coordinates $x = X/Z, y = Y/Z$ on the open set $Z \neq 0$, this curve looks like the zero locus of $y^2 - x^3$.) We have a map $\mathbb{P}^1 \rightarrow C$ given by mapping $[S, T] \mapsto [ST^2, T^3, S^3]$. In affine coordinates, this is $t \mapsto (t^2, t^3)$. This map is bijective, but at the origin, the differential is the zero map, and thus this is not an isomorphism.

21.1 Affine and projective tangent spaces

To start, we will take a more concrete example inspired by everyone's first course in calculus.

Example 21.3. Take $C := V(f(x, y))$. Then, at some point $p = (x_0, y_0)$. Then, the tangent line is the zero locus of

$$\frac{\partial f}{\partial x}(p)(x - x_0) + \frac{\partial f}{\partial y}(p)(y - y_0).$$

A priori, this zero locus is not a vector space. We need to translate to the origin (or set p to be the origin equivalently).

Definition 21.4. If $X \subset \mathbb{P}^n$, $p \in X$, define the **projective tangent space** $\mathbb{T}_p X$ to X at p to be the closure in $\mathbb{P}^n$ of affine tangent spaces to X at p in $\mathbb{A}^n \subset \mathbb{P}^n$. This is a somewhat uninspiring definition. We can do this explicitly. If $X = V(F_1, \dots, F_i)$. Then,

$$\mathbb{T}_p X = \left\{ [W_0, \dots, W_n] : \sum W_i \frac{\partial F_\alpha}{\partial Z_i}(p) = 0 \quad \forall \alpha \right\}.$$

Remark 21.5. Todo list for our future constructions:

- Gauss map
- Variety of tangent planes
- Variety of tangent lines

- Secant lines

Recall the definition from last lecture of the Gauss map: $p \mapsto \mathbb{T}_p X \in \mathbb{G}(k, n)$.

Example 21.6. Take $C = V(F) \subset \mathbb{P}^2$ where $F = F(X, Y, Z)$ is of degree d . Then, $\mathcal{G} : C \rightarrow \mathbb{P}^{2*}$ is given by

$$p \mapsto \left[\frac{\partial F}{\partial X}(p), \frac{\partial F}{\partial Y}(p), \frac{\partial F}{\partial Z}(p) \right].$$

Exercise 21.7. The Gauss map on a smooth plane curve is birational onto its image in characteristic zero.

Proposition 21.8. *If X is singular, the Gauss map is well defined and regular on the open subset $X_{sm} \subset X$, giving the rational map $\mathcal{G} : X \dashrightarrow \mathbb{G}(k, b)$.*

Definition 21.9. The image $\mathcal{T}X$ of $\mathcal{G}$ is called the variety of tangent planes to X .

Example 21.10. Take $X = V(Y^2Z - X^2(X + Z)) \subset \mathbb{P}^2$. In affine coordinates where $Z \neq 0$, we get the zero locus $y^2 - x^2(x + 1)$. This is the nodal cubic. Away from the origin, the gauss map is well defined. As we approach the origin along the curve, there are two possible tangent lines. So, much in the same way that we used blowups, the Gauss map is able to pull apart these two branches at the origin.

Now, if we take a singularity formed by two tangent curves incident at a point and take the gauss map at the point, we get a pair of transversely intersecting curves, and taking the gauss map again, we get a smooth variety. It is an open question whether every singularity can be resolved in this manner.

Definition 21.11. Recall the variety of tangent planes $\mathcal{T}X \subset \mathbb{G}(k, n)$. The **tangential variety** is

$$TX := \bigcup_{\Lambda \in \mathcal{T}X} \Lambda \subset \mathbb{P}^n.$$

Equivalently, taking our usual incidence correspondence $\Sigma := \{\Lambda, p\} : p \in \Lambda\} \subset \mathbb{G}(k, n) \times \mathbb{P}^n$ with the usual projection maps. Then, $TX = \pi_2(\pi_1^{-1}(\mathcal{T}X))$.

Exercise 21.12. What does the tangential variety of the twisted cubic look like? This is the simplest nontrivial example (as all the smaller tangential varieties are linear subspaces).

Definition 21.13. The **variety of tangent lines** to X smooth $\subset \mathbb{P}^n$ is

$$\mathcal{T}_1(X) := \{\ell \subset \mathbb{G}(1, n) : \exists p \in X, p \in \ell \subset \mathbb{T}_p X\}.$$

This is a closed subvariety of $\mathbb{G}(1, n)$.

Remark 21.14. For $X \subset \mathbb{P}^n$, we get a rational map $s : X \times X \dashrightarrow \mathbb{G}(1, n)$. The image of this map is exactly the image $s(X \times X \setminus \Delta) \cup \mathcal{T}_1(X)$

22 Lecture 22: 4/16/26

Today we are going to wrap up our discussion of tangent spaces (and possibly tangent cones).

Next week, we will introduce the degree of a projective variety.

The final lecture will be about parameter spaces, specifically Chow varieties and Hilbert Schemes. This content will not be on the final exam.

Proposition 22.1 (Lefschetz principle). *A theorem in algebraic geometry that holds over $\mathbb{C}$ in fact holds over any algebraically closed field of characteristic 0.*

Using the complex numbers can often give us more tools at our disposal, as our spaces will then have a classical topology, as well as the Zariski topology.

Remark 22.2 (Why is the Lefschetz principle true?). A statement in algebraic geometry over a field K involves finitely many varieties and maps between them, which in turn involve a finite collection of coefficients $\{c_\alpha\} \subset K$. We can then replace K by the subfield $\mathbb{Q} \subset \overline{\mathbb{Q}(c_\alpha)} = L \subset K$. Importantly, L has finite transcendence degree over $\mathbb{Q}$. So, we have an inclusion $L \subset \mathbb{C}$.

Definition 22.3 (Flexes of plane curves). Let $C = V(F) \subset \mathbb{P}^2$ is a smooth irreducible plane curve. Now, consider a line $L \subset \mathbb{P}^2$, and consider some $p \in L \cap C$. Define $m_p(L \cdot C)$ to be the order of the zero of $F|_L$ at p . If L is not the tangent line, $m_p(L \cdot C) = 1$. Otherwise, if $L = T_p C$, we have $m_p(L \cdot C) > 1$. We say a point p is a flex point of C if $m_p(T_p C \cdot C) > 2$.

Theorem 22.4. *If K has characteristic 0, then a smooth curve C of degree at least 2 can have only finitely many flexes (i.e. not every pt of C is a flex). This is not true in positive characteristic.*

Proof. By Lefschetz, we can assume that $K = \mathbb{C}$. Locally, in the classical topology, we can write C as the image of a map from $\Delta \rightarrow \mathbb{CP}^2$ sending $z \mapsto v(z) \in \mathbb{C}^3$. In these terms, we see that $z \in \Delta$ is a flex of C if and only if $v(z) \wedge v'(z) \wedge v''(z) = 0$. What if $v(z) \wedge v'(z) \wedge v''(z) \equiv 0$? Then, we can try to take the derivative. When we do this, we get

$$v'(z) \wedge v'(z) \wedge v''(z) \pm v(z) \wedge v''(z) \wedge v''(z) \pm v(z) \wedge v'(z) \wedge v'''(z) = v(z) \wedge v'(z) \wedge v'''(z).$$

We can keep differentiating and get that $v(z) \wedge v'(z) \wedge v^{(k)} \equiv 0$ for all k . Then, the conclusion is that C is the line spanned by $v(z)$ and $v'(z)$. $\square$

Theorem 22.5. *Say a line L is a bitangent line to C if there exist $p \neq q \in C$ such that $L = T_p C = T_q C$. Then, if $C \subset \mathbb{P}^2$ is a curve not a line, there exist only finitely many bitangent lines.*

22.1 Dual varieties

Assume that $X \subset \mathbb{P}^n$ is a smooth variety of dimension k . In order for any of this to make sense, assume that every irreducible component of X has dimension k .

Definition 22.6. Say a hyperplane $H \subset \mathbb{P}^n$ is tangent to X if $H \supset T_p X$ for some $p \in X \cap H$.

Definition 22.7. The **dual variety** $X^* := \{H \in \mathbb{P}^{n*} : H \text{ tangent to } X\} \subset \mathbb{P}^{n*}$.

Claim 22.8. We say X^* is a projective variety in $\mathbb{P}^{n*}$.

Proof. As should be completely expected by this point, we construct an incidence correspondence. Consider:

$$\begin{array}{ccc} & \Phi := \{(p, H) \in X \times \mathbb{P}^{n*} : H \supset \mathbb{T}_p X\} & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ X & & \mathbb{P}^{n*} \end{array}$$

Here, we have $\pi_2(\Phi) = X^*$. (One should justify to themselves that Φ is in fact a variety, using the fact that the Gauss map is regular). $\square$

Remark 22.9. Now, we can use the same setup in case that X is singular. Define $\Phi^* := \{(p, H) \in X_{sm} \times \mathbb{P}^{n*} : H \supset \mathbb{T}_p X\}$, and let Φ be the closure of Φ^* . We can then perform the same projection onto the $\mathbb{P}^{n*}$ factor.

Proposition 22.10. *This incidence correspondence also suggests that $\dim \Phi = n - 1$ and Φ is irreducible, as X is irreducible of dimension k and the fibers of the projection on X are isomorphic to $\mathbb{P}^{n-k-1}$. So, we naively expect that π_2 is generically one to one, and thus X^* has dimension $n - 1$. This is not always true.*

Definition 22.11. We say X is **deficient** if X^* has dimension less than $n - 1$ and define the deficiency $\delta(X^*) := n - 1 - \dim X^*$.

Remark 22.12. We can see from the projection that a variety is deficient iff every tangent hyperplane is tangent along a positive dimension subvariety.

Example 22.13 (Segre embedding). Take $X = \sigma(\mathbb{P}^1 \times \mathbb{P}^2) \subset \mathbb{P}^5$. This surface is given by the union of two-planes, and thus each tangent plane is tangent to a curve isomorphic to $\mathbb{P}^1$, and thus X^* has dimension 3, not 4.